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RESEARCH ARTICLE

A Multi-Criteria Framework for Selecting Machine Learning Techniques for Industrial Fault Prognosis

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ABSTRACT Selecting appropriate Machine Learning (ML) techniques for fault prognosis remains a critical yet often understructured step in developing predictive maintenance strategies for industrial systems. While numerous studies apply ML to estimate Remaining Useful Life (RUL) or forecast failure probabilities, model selection is frequently guided by *ad hoc* practices or narrow performance metrics, overlooking contextual factors such as data availability, interpretability, automation level, and deployment feasibility. This paper presents a parameterized, multi-criteria decision-making framework to support ML techniques selection in fault prognosis, particularly within energy-related applications. Derived from a structured literature survey, the framework introduces twelve selection parameters, divided into primary requirements and secondary evaluation criteria. These parameters, such as label requirements, model complexity, and transferability, allow users to eliminate unsuitable techniques and rank viable candidates according to application-specific constraints. The framework is applied to a real-world use case involving failure prediction in electrical substations, illustrating how it supports transparent, replicable, and operationally grounded model selection. Results demonstrate the framework's adaptability to different industrial contexts and its relevance for decision-making in energy systems. By bridging empirical insights with implementation demands, the proposed approach offers a practical tool for aligning ML technique selection with the goals of energy-sector prognostics and maintenance planning.

INDEX TERMS Evaluation methods, industrial systems, machine learning, predictive maintenance, prognostics, remaining life assessment.

I. INTRODUCTION

Industrial systems increasingly operate under demanding conditions, where unplanned failures can trigger significant financial losses, safety risks, and cascading operational disruptions. To mitigate such events, maintenance strategies have evolved from reactive and scheduled interventions to condition-based approaches, grounded in continuous monitoring and early detection. Within this context, fault prognosis plays a critical role by anticipating degradation and informing preventive actions before failures occur [1].

Unlike fault detection or diagnosis, prognosis is not limited to identifying anomalies or causes. It entails a structured process: acquiring sensor data, constructing Health Indicators (HIs) that represent system condition, and forecasting degradation trajectories. These elements support the timely identification of emerging risks and enable proactive, reliability-centered decision-making [2], [3].

Recent advances in Machine Learning (ML) have significantly expanded the capabilities of prognostic models. Data-driven techniques can capture nonlinear degradation

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dynamics, reduce dependence on handcrafted features, and adapt to diverse system behaviors. In high-variability environments, such as wind farms or large power plants, AI-based models have proven effective in extracting patterns from multivariate time series, Supervisory Control and Data Acquisition (SCADA) systems, and other monitoring sources. Nonetheless, their deployment in real-world settings is often constrained by the scarcity of labeled data, high computational demands, and limited model transparency, particularly critical in energy and safety-sensitive domains [4], [5].

Despite growing interest in data-driven fault prognosis, no established methodology currently guides the selection of ML techniques across prognostic scenarios. Comparative studies often prioritize accuracy or runtime, neglecting essential factors such as data availability, interpretability, automation level, operational feasibility, and alignment with operational constraints. Even in multi-algorithm reviews or RUL estimation studies, systematic guidance for model selection remains rare [6], [7], [8]. Broader surveys confirm that selection is frequently based on *ad hoc* experimentation or user familiarity rather than transparent, evidence-based criteria [9], [10].

This gap is further reflected in the dominance of benchmark-centric studies that emphasize metrics like accuracy, Root Mean-Square Error (RMSE), Mean Absolute Percentage Error (MAPE), or F1-score, while overlooking upstream concerns related to model suitability or deployment feasibility [11], [12], [13]. Even when industrial datasets are used, model choice is rarely tied to generalizable criteria such as interpretability, computational footprint, or the nature of the degradation process [14], [15], [16].

Some recent works have made progress by comparing deep learning and explainable AI models. However, their scope is often restricted to performance rankings, offering limited insight into broader selection criteria, such as data sparsity, domain specificity, or integration with industrial workflows [17], [18]. Similarly, application-specific reviews typically organize algorithm comparisons by output metrics, without establishing systematic links between model characteristics and contextual needs [7], [8].

A smaller group of studies has begun advocating for structured and theoretically informed approaches. Arena et al. [19] propose a conceptual framework that considers user context and data properties but lacks operational selection rules. Rocchetta et al. [20] apply scenario theory to assess robustness under uncertainty, while Qiao et al. [21], Cheng et al. [4], and Coronel et al. [10] classify ML techniques by system type or failure mode, but stop short of operationalizing selection procedures.

Some studies attempt to bridge benchmarking with conceptual insight. Bracale et al. [22] exemplify this by comparing models while analyzing assumptions in lithium-ion battery prognosis. Fernandes et al. [23] and Amaya-Sanchez et al. [24] incorporate expert feedback and usability assessments in industrial case studies, with the latter focusing specifically on power generation systems. Yet in most of these cases, model selection is retrospective, driven by *post hoc* performance rather than guided by predefined evaluation criteria.

Taken together, these findings highlight a persistent gap: although both empirical and theoretical works acknowledge the complexity of technique selection in fault prognosis, few offer structured, transparent, and generalizable procedures.

Rather than evaluating individual model performance, this study aims to fill that gap by providing a structured tool to support the preliminary selection of ML techniques in industrial prognosis settings. The proposed approach is intended to guide decision-making under practical constraints, not to implement or validate full predictive pipelines.

This paper addresses that gap by introducing a parameter-driven framework for selecting ML techniques tailored to fault prognosis. The approach departs from traditional benchmarking by integrating theoretical, computational, and contextual factors into a unified, replicable workflow. Twelve selection parameters, proposed in this work, are structured into a two-phase decision process: six serve to eliminate incompatible techniques, while four support a Multi-Criteria Decision-Making (MCDM) procedure to rank viable options.

The final two parameters serve as contextual attributes for post-selection analysis. This structure supports context-sensitive and evidence-based decision-making, offering a practical alternative to heuristic or accuracy-centric practices.

The remainder of this article is structured as follows: Section II outlines the foundations of fault prognosis, distinguishing it from related tasks and detailing the main modeling stages. Section III presents the survey methodology used to identify and justify the selection parameters. Section IV introduces the proposed framework and decision pipeline. Section V applies the framework to a real-world energy system. Section VI concludes the article and suggests directions for future research.

II. FOUNDATIONS OF FAULT PROGNOSIS IN INDUSTRIAL SYSTEMS

Fault prognosis is a key component of modern Reliability-Centered Maintenance (RCM) frameworks, working in tandem with fault detection and diagnosis to provide early and actionable insights into system degradation. Although interconnected, these tasks are technically distinct. Fault detection identifies deviations from normal operation without determining their cause. Fault diagnosis investigates the nature, location, and origin of the anomaly. Fault prognosis, in contrast, focuses on forecasting future fault evolution, typically by estimating Remaining Useful Life (RUL) or Failure Probability within a Predefined Horizon (FPPH) [1], [25]. These capabilities are particularly valuable in energy systems, where unplanned failures in assets such as transformers or turbines can have cascading effects.

Detection often relies on anomaly detection, while diagnosis uses expert systems or classification models. In classification-based implementations, both tasks may be combined into a single predictive layer [26], [27]. Prognosis, however, requires distinct modeling capabilities to capture degradation dynamics and the temporal evolution of system health [4], [20], [25].

The fault prognosis process can be formally structured into four technical stages [3], [21]:

- Data acquisition;
- Health Indicator (HI) construction;
- Health Stage (HS) division;
- Degradation Trend Prediction (DTP).

The first stage involves acquiring operational and condition-monitoring data, such as vibration, temperature, current, or environmental readings, from sensors installed on critical components. Although fundamental, data acquisition often receives less emphasis in the literature, as many industrial systems prioritize control, protection, or diagnostics rather than prognosis.

In the second stage, HIs are built to extract degradation-relevant information from raw data. Effective HIs should be sensitive to degradation, monotonic over time, and robust to noise [3]. While traditional pipelines construct HIs explicitly, deep learning models can capture degradation patterns implicitly through learned representations, even if not explicitly designed for that purpose [21].

Choosing between explicit HIs and direct end-to-end learning has practical implications. End-to-end models simplify deployment but tend to be less interpretable and more prone to overfitting, especially under noise or early-stage degradation. Conversely, HI-based approaches often yield greater stability and generalizability, particularly when enhanced by autoencoders or signal decomposition [3], [21].

Techniques for HI construction include linear methods, e.g., Principal Component Analysis (PCA) [28] statistical and spectral feature extraction [2], [3], and deep embeddings from representation learning models [14]. Once defined, HIs form the foundation for downstream tasks.

The third stage, HS division, segments the HI trajectory into discrete regimes (e.g., healthy, degraded, critical) via rule-based partitioning, clustering, or supervised classification. This enhances interpretability, supports tailored modeling per regime, and facilitates estimation of state-dependent failure probabilities [3]. It also enables linking degradation levels to maintenance policies. This is especially relevant in critical energy infrastructure, where condition-based maintenance policies must comply with regulatory and operational reliability standards.

The final stage involves predicting degradation outcomes, such as RUL or FPPH. This may be done by projecting HIs toward a failure threshold using regression, classification, or time-series forecasting models [2], [14], [21]. Parametric models fit analytical curves (e.g., linear or exponential) to estimate RUL [28], while stochastic processes like Wiener or Gamma models offer probabilistic failure forecasts, even in the absence of labeled failures [3]. ARIMA and related models extrapolate future HI values based on past observations [2].

Across all methods, prognosis depends on a critical HI threshold indicating system failure, defined by physical criteria, expert input, or statistical analysis [3]. Deep learning approaches such as Long Short-Term Memory (LSTM) and Temporal Convolutional Networks (TCNs) excel at modeling complex temporal dependencies, though they require labeled RUL data [14], [26]. To reduce this dependency, semi-supervised methods, such as autoencoders tracking reconstruction errors, have been proposed [29].

Fully unsupervised approaches that detect degradation solely from HI dynamics, without labels or thresholds, remain under development. Though promising conceptually, they still face limitations in reliability, explainability, and alignment with risk-based maintenance strategies [2].

Fault prognosis strategies are generally categorized as model-based, data-driven, or hybrid [3], [25]. Model-based methods rely on physical laws or first-principles modeling [1], [25], while hybrid methods combine physical models with data-driven insights [3], [25]. Although valuable in domains with well-understood degradation mechanisms, these approaches often require detailed system knowledge and offer limited flexibility in adapting to different operating conditions.

Data-driven approaches, in contrast, are more adaptable, scalable, and suited to complex or partially understood systems. They thrive in industrial environments, particularly in energy applications, where detailed physical models are often unavailable, incomplete, or difficult to maintain over time. These models can be updated continuously and support rapid adaptation to changing conditions [14], [26].

For these reasons, this article focuses exclusively on data-driven fault prognosis. The analysis follows the four-stage structure described above, i.e., data acquisition, HI construction, HS division, and DTP, and explores how ML techniques are applied at each stage. In addition to modeling tasks (e.g., regression, classification, time series), selection also considers data availability, interpretability, and computational constraints. The next section presents the structured literature synthesis, laying the foundation for the parameterized framework introduced in Section IV.

III. SURVEY OF MACHINE LEARNING TECHNIQUES FOR FAULT PROGNOSIS

In industrial settings, the adoption of predictive algorithms depends not only on performance but also on maturity, interpretability, and demonstrated success in real-world applications [30]. Accordingly, this section does not present a conventional systematic literature review in the academic sense, as that is not the purpose of this article. Instead, it aims to identify and classify machine learning techniques that have been effectively applied to fault prognosis in engineering systems, either at scale or in credible industrial case studies,

based on a curated sample of articles indexed in one of the most reputable scientific repositories.

The focus is both systematic and application-oriented. The goal is to establish a taxonomy of techniques that surpass a minimum threshold of industrial maturity and relevance. This aligns with the observations of Hoffmann et al. [30], who highlight that the viability of predictive maintenance strategies hinges on algorithmic robustness, transparency, and compatibility with operational constraints.

To this end, a five-stage research protocol, summarized in Figure 1, was developed and implemented between August 2023 and January 2024. The protocol prioritized the identification of mature techniques using structured metadata and high-quality indexing. All searches were conducted exclusively through the Web of Science Core Collection.

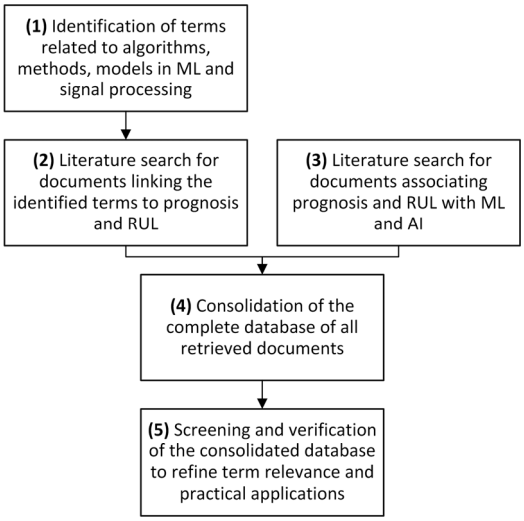


FIGURE 1. Schematic representation of the research protocol.

The decision to restrict the search to the Web of Science was deliberate. This database offers structured metadata, standardized indexing, and rigorous curation across engineering and applied science journals, features essential for reproducibility, bibliometric analysis, and filtering of non-peer-reviewed or low-relevance content. Since the objective was not to exhaust the academic literature but to identify techniques with proven industrial relevance, other databases were considered unnecessary. A single, high-quality database was deemed sufficient to support the construction of a reliable, industry-aligned taxonomy.

The next subsections detail each stage of the research protocol, including procedures, search strategies, and inclusion criteria. This step-by-step account also clarifies the rationale for the structure of the selection framework introduced in Section IV.

A. IDENTIFICATION OF TERMS RELATED TO ALGORITHMS, METHODS, AND MODELS IN ML AND SIGNAL PROCESSING

The first stage of the research protocol involved constructing a comprehensive vocabulary of techniques used in fault prognosis. Rather than adopting predefined taxonomies from previous reviews, a bottom-up approach was employed. A total of 178 terms were identified through an unstructured exploratory search, guided by the authors' technical expertise and prior knowledge of the domain. Terms were extracted based on their explicit presence in metadata fields, i.e., titles, abstracts, and keywords, of relevant publications.

The resulting vocabulary spans a wide range of machine learning and signal processing techniques, including classical models (e.g., Support Vector Machines, Decision Trees), deep learning architectures (e.g., Convolutional Neural Networks, Long Short-Term Memory), probabilistic models (e.g., Kalman Filters, Hidden Markov Models), and signal processing methods (e.g., Wavelet Transform, Principal Component Analysis).

Selection followed an iterative process that combined expert input, literature cross-referencing, and exploratory queries in academic databases. Both full expressions and their acronyms (e.g., "Support Vector Machine" and "SVM") were included when relevant.

In addition to learning algorithms, feature extraction and signal processing techniques were included due to their common use in industrial prognosis. This broader scope ensures the study accounts for both predictive models and essential preprocessing methods. The complete list of identified terms is provided in Appendix (Table 15).

B. LITERATURE SEARCH FOR DOCUMENTS LINKING THE IDENTIFIED TERMS TO PROGNOSIS AND RUL

The second stage involved querying each of the 178 identified terms individually in the Web of Science Core Collection, using structured search expressions that combined the term (or its acronym) with keywords related to fault prognosis. Each query followed the general structure: "[term]" OR "[acronym]" AND ("prognosis" OR "prognoses" OR "prognostics" OR "remaining useful life" OR "RUL") AND "fault".

Searches were restricted to the Topic field, covering titles, abstracts, and keywords, to ensure thematic relevance and minimize semantic noise. When acronyms could lead to ambiguity (e.g., "DBN", which may refer to Deep Belief Network, Dynamic Bayesian Network, or other meanings), only the full term was used to preserve search precision.

Of the 178 queries, 122 returned at least one relevant result, yielding an initial pool of 3,225 documents. After removing duplicates, i.e., records retrieved in multiple queries, the dataset was reduced to 1,853 unique documents.

A frequency table was constructed to indicate how often each term appeared in the metadata of the retrieved documents. This served as a preliminary proxy for the relative prominence of each technique in the fault prognosis literature. The complete table is available in Appendix (Table 16). Notably, the five most frequently cited techniques were:

1) Wavelet Transform: 295 records.
2) Convolutional Neural Networks (CNN): 213 records.

3) Long Short-Term Memory (LSTM): 205 records.
4) Principal Component Analysis (PCA): 185 records.
5) Support Vector Machines (SVM): 185 records.

C. LITERATURE SEARCH FOR DOCUMENTS ASSOCIATING PROGNOSIS AND RUL WITH ML AND AI

While the previous step targeted documents referencing specific techniques, this third stage broadened the scope to include publications that address fault prognosis from a conceptual or system-level perspective. The objective was to capture studies that, even without citing algorithms in their topic metadata, contribute meaningfully to the methodological discussion around prognosis and RUL estimation using machine learning or artificial intelligence.

To that end, a single structured query was formulated: (‘‘prognosis’’ OR ‘‘prognoses’’ OR ‘‘prognostics’’ OR ‘‘remaining useful life’’ OR ‘‘RUL’’) AND ‘‘fault’’ AND (‘‘machine learning’’ OR ‘‘artificial intelligence’’).

This expression was designed to maximize coverage of data-driven prognostic studies and ensure that architectural, integrative, or application-focused works were not overlooked due to the absence of algorithm-specific terms. The search was applied to the Topic field of the Web of Science Core Collection, maintaining the same field constraints previously used.

The query returned 388 additional documents, which were incorporated into the working corpus before consolidation. By including this complementary set, the protocol ensured that the final dataset would reflect not only the prevalence of individual techniques but also the broader spectrum of methodological contributions to ML-based fault prognosis.

D. CONSOLIDATION OF THE COMPLETE DATABASE OF ALL RETRIEVED DOCUMENTS

The results from Sections B and C were merged into a unified corpus. After removing duplicate entries, i.e., documents retrieved by more than one query, the final dataset comprised 1,982 unique records.

This consolidated set comprises only peer-reviewed publications, including journal articles, conference papers, and book chapters, spanning the domains of engineering, artificial intelligence, reliability, and industrial automation.

By combining both technique-specific and broader methodological studies, this step ensured that the resulting corpus captures not only the diversity of algorithms but also the breadth of applications in machine learning-based fault prognosis. Although not exhaustive, the dataset represents a robust and representative sample of current research, supporting the bibliometric and methodological analyses presented in the next sections.

E. SCREENING AND VERIFICATION OF THE CONSOLIDATED DATABASE TO REFINE TERM RELEVANCE AND PRACTICAL APPLICATIONS

The final step of the protocol involved a qualitative screening of the consolidated corpus to ensure alignment with the study’s focus on machine learning-based fault prognosis. Although all documents had been retrieved through structured queries, not all met the inclusion criteria upon closer examination. The screening process involved manual review of titles, abstracts, and keywords, focusing on two main aspects:

- Whether the study addressed prognostic modeling or RUL estimation, rather than solely fault detection, diagnosis, or classification;
- Whether the cited technique was effectively implemented, analyzed, or discussed within a concrete prognostic context.

Documents that failed to meet these criteria were excluded. Review articles were retained only when they explicitly described how the techniques were applied within prognostic frameworks. In cases where metadata was inconclusive, a full-text review was conducted to assess relevance.

As a result of this process, the final corpus consisted of 984 peer-reviewed documents, forming the empirical basis for the frequency analysis, co-occurrence mapping, and clustering presented in the next section. Across this corpus, 253 unique techniques were identified through explicit mentions in metadata fields. This extended list is presented in Appendix (Table 17) and reflects the methodological diversity observed in the literature.

To enhance analytical robustness, only techniques cited in at least 10 distinct documents were retained for further analysis. This threshold balances coverage and relevance, excluding rarely used or less mature methods while preserving methodological variety. A total of 40 techniques met this criterion (highlighted in bold in Table 17).

Table 1 illustrates the most frequently cited techniques, listing the top 22 entries from the final list, i.e., those cited

TABLE 1. Frequency of terms and acronyms identified in the literature on fault prognosis.

Technique Name	Acronym	Frequency
Wavelet	-	144
Long Short-Term Memory	LSTM	132
Convolutional Neural Network	CNN	104
Particle Filter	-	91
Support Vector Machine	SVM	80
Principal Component Analysis	PCA	78
Kalman Filter	KF	62
Clustering	-	53
Hidden Markov Models	HMM	53
Support Vector Regression	SVR	50
Anomaly Detection	-	49
Recurrent Neural Network	RNN	48
Transfer Learning	-	44
Autoencoder	-	42
Memory Networks	-	31
Bayesian Network	-	26
Brownian Motion	-	26
Knowledge-Based Expert System	KBES	26
Self-Organizing Map	SOM	26
Deep Neural Network	DNN	25
Fuzzy Logic	-	25
Gaussian Process Regression	GPR	25

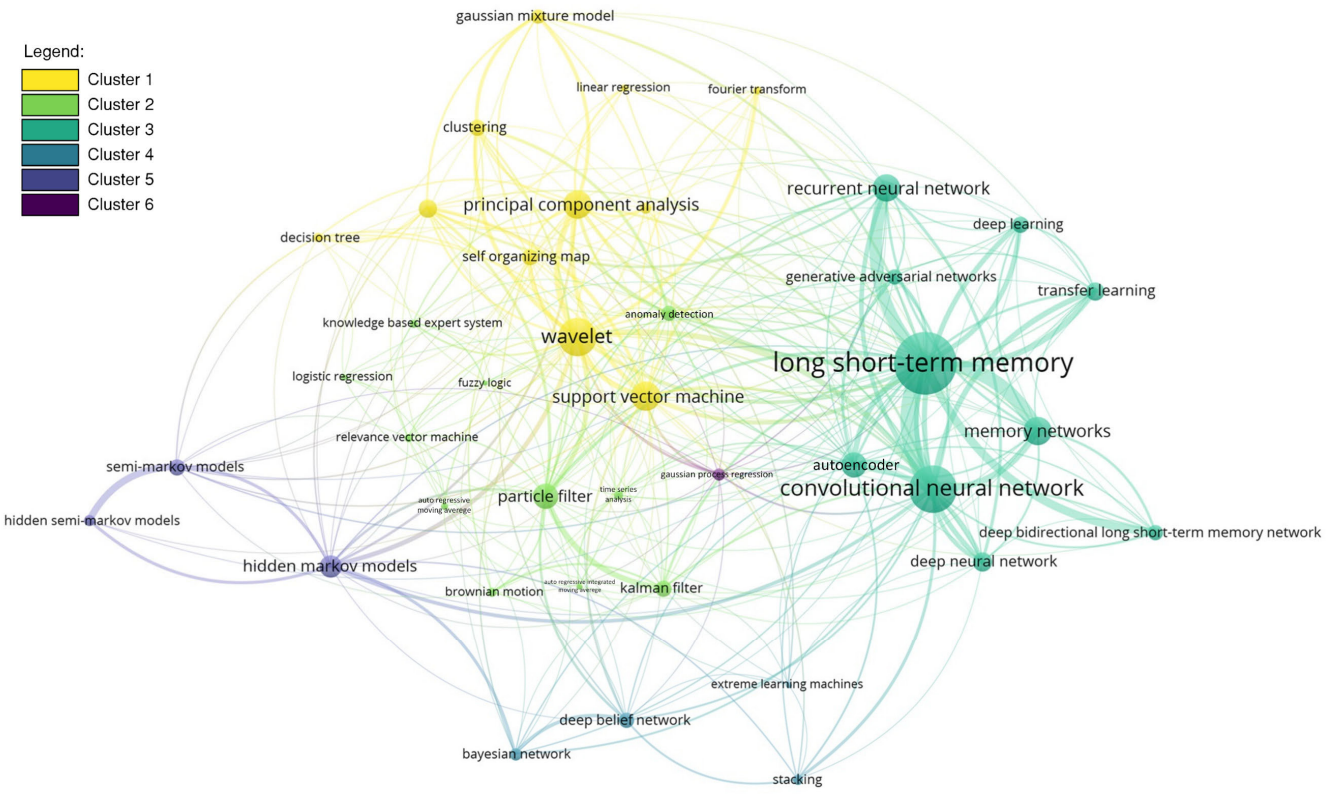


FIGURE 2. Co-occurrence Map of the 40 most frequent techniques.

in at least 25 documents, along with their acronyms (where applicable) and the number of documents in which each technique was actively applied to fault prognosis, as verified during screening.

F. SURVEY RESULTS

This section presents the main findings of the survey, focusing on the structural relationships among the 40 most frequently applied techniques in fault prognosis. In addition to identifying co-occurrence patterns and methodological groupings, the analysis explores how these techniques are typically applied within predictive maintenance frameworks, offering practical insight into their functional roles.

Based on the 40 most frequent techniques identified in the previous stage, a set of bibliographic relational maps was constructed using VOSviewer, a free software tool for creating and visualizing bibliometric networks [31].

Figure 2 displays a co-occurrence map of the selected terms, restricted to those cited in at least 10 distinct documents. In this visualization, node size and label font represent term frequency and connectivity, while edge thickness reflects the strength of co-occurrence between techniques. Clusters were formed based on high internal association using VOSviewer's modularity-based algorithm.

To ensure analytical rigor and reproducibility, the analysis applied the association strength method to normalize co-occurrence frequencies by accounting for individual term prevalence. The layout followed a force-directed placement model with default parameters (Attraction = 2, Repulsion = 0), optimized over 1000 iterations with an initial step size of 1.0, reduced progressively by a factor of 0.75 until reaching a convergence threshold of 0.001. A single random start and fixed seed (0) were used to ensure deterministic layout generation.

Clustering was performed with resolution set to 1.0 and a minimum cluster size of 1. To enhance robustness, the clustering process was repeated across 10 random starts, each with 10 iterations, also using a fixed random seed (0). These settings produced six distinct clusters that reflect dense empirical associations among techniques frequently co-applied in fault prognosis studies. The result offers a data-driven basis for downstream classification and the derivation of selection attributes, while emphasizing that these associations represent statistical affinities rather than causal relationships.

The resulting clusters and their constituent techniques, along with summaries of their methodological characteristics, are presented in Table 2. These clusters capture underlying patterns related to data modality (e.g., temporal vs. static),

TABLE 2. Clusters of machine learning techniques and their characteristics.

Cluster	Techniques	Cluster Characteristics
1	Clustering, Decision Tree, Fourier Transform, GMM, Linear Regression, PCA, Random Forest, SOM, SVM, SVR, Wavelet	Foundational techniques used across domains. Includes supervised models (e.g., Decision Tree, Linear Regression), unsupervised methods (e.g., Clustering, GMM), and dimensionality reduction tools (e.g., PCA). Known for interpretability and efficiency, but with limited support for temporal modeling or deep architectures.
2	Anomaly Detection, ARIMA, ARMA, Brownian Motion, Fuzzy Logic, Kalman Filter, KBES, Logistic Regression, Particle Filter, RVM, Time Series Analysis	Time-series and anomaly detection models emphasizing temporal dynamics. Often used for sequential data modeling across domains, contrasting with the more static or deep-learning approaches in other clusters.
3	Autoencoder, CNN, BiLSTM, Deep Learning Architectures, DNN, GAN, LSTM, Memory Networks, RNN, Transfer Learning	Advanced deep learning techniques supporting hierarchical and representation-based learning. Frequently applied in complex and data-rich prognosis scenarios, reflecting the shift toward end-to-end modeling.
4	Bayesian Network, DBN, ELM, Stacking	Probabilistic and ensemble approaches. Typically applied in niche or hybrid settings, where uncertainty modeling or aggregation across models is required.
5	HMM, HSMM, SMM	Markovian models for discrete state transitions. Well-suited to sequential processes where system evolution depends solely on the current state.

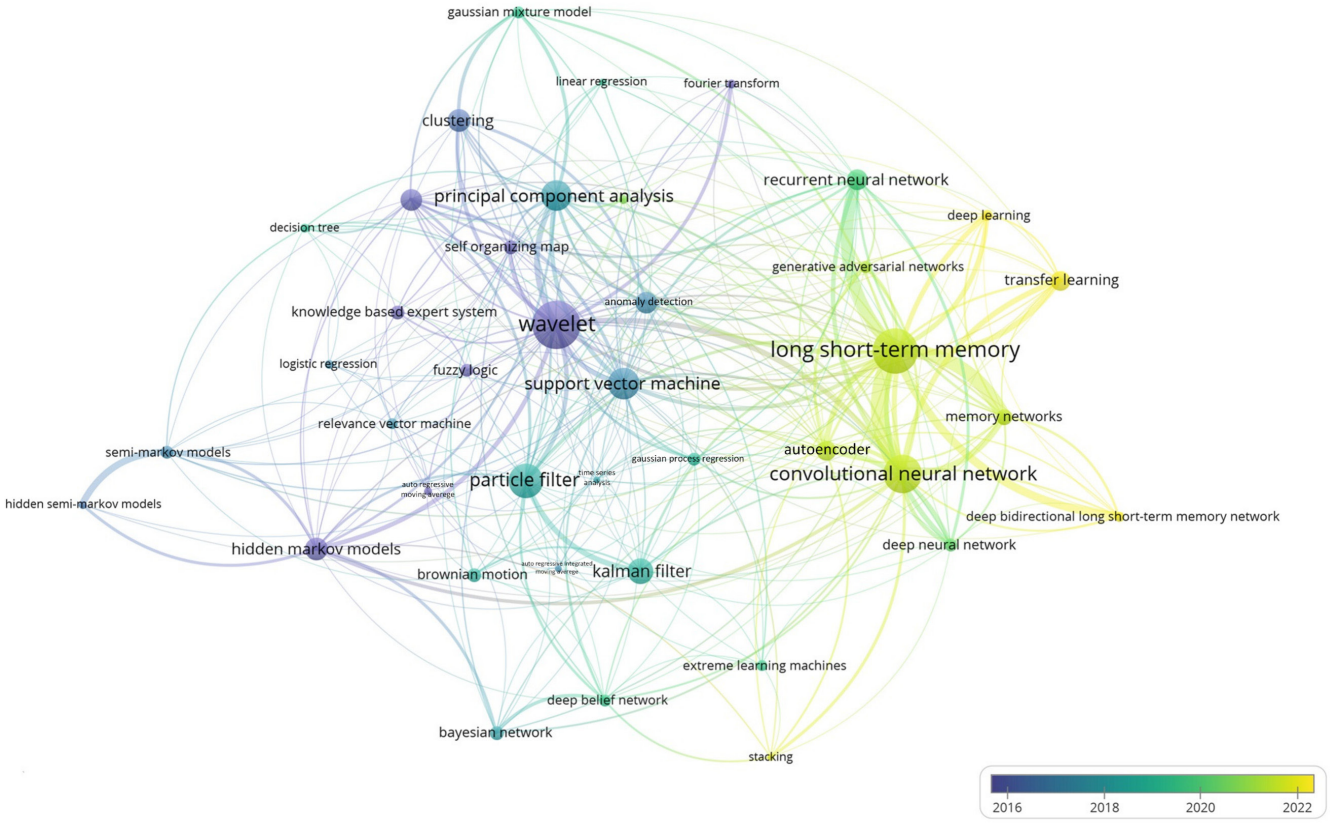


FIGURE 3. Techniques temporal overlay of technique adoption (average publication year).

methodological orientation (e.g., probabilistic, deep learning, feature-based), and application scope (e.g., general-purpose vs. specialized techniques).

Figure 3 incorporates temporal information into the co-occurrence map by coloring each node based on the average publication year of its associated documents. While node size and font continue to represent term frequency, the color gradient, from purple (older) to bright yellow (more recent), indicates the relative recency of each technique's adoption.

The map reveals a clear temporal stratification of modeling paradigms. Older techniques, such as PCA, Wavelet Transform, SVM, and HMM (in darker tones), occupy central or peripheral regions of early research clusters and are typically associated with feature-based modeling and probabilistic estimation. In contrast, recent methods like LSTM, CNN, Memory Networks, GAN, and Transfer Learning (in bright yellow) dominate post-2020 literature, evidencing the increasing prevalence of deep learning in prognostics.

Techniques in intermediate shades often serve as methodological bridges. Kalman Filter, for example, remains central due to its historical importance in temporal modeling and persistent relevance. Autoencoder and Stacking occupy transitional positions, suggesting their hybrid use in pipelines that combine feature extraction, dimensionality reduction, and ensemble strategies.

Additionally, some older models, such as Particle Filter and Bayesian Network, maintain strong links to recent approaches, underscoring their continued applicability in scenarios that require probabilistic reasoning and uncertainty management.

Altogether, Figure 3 provides empirical evidence of the field's evolution: from signal-processing and statistical modeling toward more data-driven, neural-based architectures. This progression reinforces the need for flexible and context-sensitive selection frameworks capable of accommodating both mature, industry-proven models and emerging techniques that leverage modern computing paradigms.

Together, the co-occurrence analysis (Figure 2) and the temporal overlay (Figure 3) provide complementary perspectives on the evolution and interrelationships among the 40 most frequent techniques in fault prognosis. The cluster map reveals empirical groupings based on shared usage patterns, while the temporal visualization highlights methodological shifts.

To consolidate these insights, Table 18 (Appendix) presents a structured synthesis that integrates three key dimensions: (i) cluster assignment derived from the co-occurrence network, (ii) prognosis stage coverage as defined in Section II, and (iii) qualitative summaries of typical applications and model configurations. This integrative view supports a more nuanced understanding of each technique's role, contextual suitability, and modeling paradigm.

While cluster assignments were directly obtained from the VOSviewer output (Table 2), the other two dimensions required interpretive analysis grounded in qualitative coding. Prognosis stages were annotated through inductive coding of each technique's function in the reviewed studies, using the three-phase taxonomy introduced in Section II as a reference. For example, techniques were classified under ''HI construction'' when used to extract or transform features that reflect system degradation, and under ''DTP'' when directly applied to forecast RUL or degradation trends.

Application summaries were derived through sentence-level content analysis, focusing on recurring modeling roles (e.g., encoder, classifier, regressor), data assumptions (e.g., temporal dependencies, need for labels), and architectural patterns (e.g., hybrid models, attention mechanisms). To ensure consistency, techniques were analyzed across multiple studies, and classifications were assigned based on repeated usage contexts rather than isolated cases.

Although this process was inherently interpretative, efforts were made to maintain internal coherence and transparency. References were curated to reflect core contributions, prioritizing studies that clearly documented implementation details and contextual applications. This methodology provides a structured yet flexible synthesis aligned with the varied nature of fault prognosis literature.

Notably, the analysis also points to a systematic gap: the data acquisition stage remains unaddressed by any of the reviewed techniques, underscoring a persistent blind spot in the literature.

IV. PROPOSED FRAMEWORK

The structured survey presented in the previous section revealed substantial diversity in the machine learning techniques applied to fault prognosis. These techniques vary considerably in terms of methodological foundations, data assumptions, interpretability, and levels of automation. Such heterogeneity underscores the importance of adopting a principled and context-sensitive approach to method selection, one that aligns model characteristics with the specific constraints and objectives of each industrial application.

To meet this need, a parameterized decision-making framework is introduced to support the selection of appropriate techniques in a structured and reproducible manner. Rather than relying on rigid taxonomies or benchmark-based rankings, the framework draws upon patterns identified in the literature review, ensuring that the selection process reflects both the technical distinctions among methods and the practical considerations of real-world deployment.

The remainder of this section details the framework's construction in five steps: derivation of selection criteria, formal definition of attributes, attribute classification and coding, functional classification, and a two-phase decision pipeline for technique selection.

A. DERIVATION OF SELECTION CRITERIA FROM SURVEY RESULTS

The twelve selection attributes proposed in this framework were not drawn from preexisting taxonomies or standardized guidelines. Instead, they were inductively derived from a systematic analysis of how machine learning techniques are actually applied in fault prognosis tasks, as documented in the reviewed literature. This derivation focused primarily on the qualitative examination of technique usage across application contexts, enabling the identification of recurring implementation patterns, modeling assumptions, and practical constraints.

While the structural mapping performed in Section III, such as co-occurrence analysis, temporal overlays, and cluster assignments, provided important contextual insights, these elements served a complementary function. They helped position each technique within the broader methodological landscape and highlighted empirical affinities, but the core attributes themselves emerged from content-level interpretation of application details reported in the studies.

Specifically, the reviewed documents were analyzed to identify what kinds of data each technique typically consumes, how much supervision is required, whether methods are adapted or reused across systems, and what levels of

interpretability or automation are generally observed in practice. These insights were synthesized to define a set of attributes that (i) reflect common trade-offs encountered in real-world deployments, (ii) allow systematic comparison of candidate methods, and (iii) support transparent and reproducible decision-making.

As detailed in Table 18, deep learning models such as LSTM, CNN, and GAN tend to appear in supervised scenarios with abundant sequential data, often requiring fully labeled training sets and offering limited interpretability. In contrast, techniques like PCA, SVM, and Random Forest are typically found in interpretable pipelines involving structured sensor logs and handcrafted features. Probabilistic filters and state-space models, including the Kalman Filter and Particle Filter, are frequently deployed when uncertainty quantification or real-time state estimation is needed.

These observed patterns informed the formulation of twelve attributes, which together provide a multidimensional profile of each technique. The attributes serve two main purposes within the framework: first, as a basis for filtering techniques that fail to meet essential feasibility constraints; and second, as a structure for ranking viable candidates according to application-specific priorities. This dual function enables the framework to align technical characteristics with contextual demands, thereby supporting more informed and justifiable method selection.

B. MACHINE LEARNING ATTRIBUTES DEFINITION FOR FAULT PROGNOSIS

This subsection defines the twelve attributes proposed in this study to characterize ML techniques for fault prognosis, focusing on feasibility, modeling behavior, and deployment suitability for structured comparison and selection.

1) LABEL REQUIREMENT (LR)

Label Requirement (LR) defines the degree to which a technique relies on labeled data during training and deployment, based on its actual use in fault prognosis applications as reported in the literature, rather than solely on its theoretical learning paradigm:

- Fully Required (Fr): Depends on fully labeled datasets, typically used in supervised settings such as RUL prediction, where time-to-failure or degradation states are known;
- Partially Required (Pr): Functions effectively with partially labeled data, aligning with semi-supervised approaches commonly adopted when some label information is available but incomplete;
- Not Required (Nr): Operates without any labeled data, relying solely on patterns extracted from unlabeled inputs, typical in unsupervised applications.

2) DATA TYPE (DT)

Data Type (DT) specifies the structure and format of the input data required or best handled by the technique. In fault prognosis, data sources commonly include:

- Structured data (Sd): Tabular sensor logs or SCADA records with predefined variables.
- Sequential data (Sq): Time series capturing temperature, pressure, or vibration evolution over time.
- Unstructured data (Un): Thermal images, spectrograms, or technician notes.
- Hybrid data (Hy): A combination of multiple formats.

Techniques differ in their ability to process these types. For example, RNNs excel in sequential data modeling, while CNNs may handle both structured and unstructured inputs when appropriately transformed. Ensuring compatibility between data structure and technique assumptions is a primary determinant in the selection process.

3) DATA AVAILABILITY (DA)

Data Availability (DA) describes the volume and quality of historical data required for the technique to perform effectively. In fault prognosis, this attribute is critical due to the limited availability of failure examples in some systems (e.g., hydrogenerator units and large synchronous motors) and the typically long degradation histories (e.g., power transformers or high-voltage circuit breakers).

- Data-Scarce Tolerant (Sc): Techniques such as probabilistic filters or domain-adaptive models that can operate reliably with minimal historical data;
- Moderate Data Requirement (Mr): Techniques that may require some labeled degradation trajectories or structured feature sets;
- Data-Rich Required (Rr): Techniques, especially deep learning models, that demand large, labeled datasets to generalize effectively.

Selecting techniques compatible with the available data volume and quality is essential to avoid underperformance or model overfitting.

4) AUTOMATION LEVEL (AL)

Automation Level (AL) indicates the degree of manual intervention required during the development, configuration, and maintenance of the technique.

- Manual (Mn): Techniques that require expert-driven feature engineering, parameter tuning, and model design. They offer high interpretability and customizability but present limitations in scalability and consistency across deployments.
- Semi-Automated (Sa): Techniques that incorporate partial automation, such as algorithmic feature selection, assisted hyperparameter tuning, or prebuilt model architectures. These are commonly applied using standard libraries and toolkits, operate within partially automated modeling pipelines, and strike a balance between expert supervision and implementation efficiency. While they may require some preprocessing or user configuration, they do not demand constant manual intervention during training or inference.

- Fully Automated (Fa): Methods that include AutoML frameworks and end-to-end deep learning solutions, aiming to minimize human involvement entirely. They support automatic data ingestion, feature processing, model selection, and retraining, making them suitable for large-scale, multi-asset environments where rapid deployment and maintenance are essential.

The automation level directly affects implementation cost, scalability, and maintainability in industrial prognosis applications.

5) TRANSFERABILITY (TR)

Transferability (TR) refers to the extent to which a model trained in one operational context can be reused or adapted to other systems, components, or environments without retraining from scratch.

- Limited Transferability (Lm): Techniques that are tightly coupled to specific data distributions, operational regimes, or sensor configurations. They require full retraining for each new application.
- Adaptable Techniques (Ad): Methods that can be re-parameterized or lightly fine-tuned when applied to similar systems, often requiring partial retraining or integration of contextual metadata.
- Highly Transferable (Ht): Techniques designed for cross-domain generalization, including those based on domain adaptation, transfer learning, or invariant feature representations. These models support seamless reuse across assets with minimal adjustment.

Transferability directly affects deployment costs, model lifespan, and scalability across heterogeneous industrial environments.

6) GENERALITY VS. SPECIALIZATION (GS)

Generality vs. Specialization (GS) distinguishes whether a technique is designed for broad applicability across diverse systems and failure mechanisms or optimized for a narrow class of problems and operational contexts.

- Specialized Techniques (Sp): Developed or tuned for specific domains, fault types, or signal characteristics. These models tend to perform well under tightly constrained conditions but may not generalize effectively beyond their original context.
- Moderately General Techniques (Md): Applicable across a range of settings but often require adjustments in preprocessing, feature design, or training configuration. They offer a balance between adaptability and performance.
- General-Purpose Models (Gp): Architectures and algorithms that have been successfully applied across multiple industries and degradation modes with minimal customization. Their versatility supports wide adoption and integration.

This attribute informs decisions where reuse, extensibility, or cross-system integration is a priority.

7) INTERPRETABILITY (IN)

Interpretability (IN) assesses how easily the outputs, structure, and reasoning of a technique can be understood by engineers, analysts, or decision-makers. Interpretability is essential in industrial settings where predictions must support transparent, auditable, and actionable maintenance decisions.

- Low Interpretability: Techniques that operate as black boxes, offering limited insight into the reasoning behind predictions without *post hoc*analysis.
- Medium Interpretability: Models that allow partial understanding of internal processes or can be interpreted using auxiliary methods such as explanation frameworks or surrogate models.
- High Interpretability: Techniques whose decision logic is directly accessible and explainable, allowing for traceable justifications of outcomes.

Interpretability directly influences user trust, regulatory compliance, and the integration of model outputs into operational workflows.

8) TRANSPARENCY AND EXPLAINABILITY (TE)

Transparency and Explainability (TE) refers to the extent to which a model's internal operations and decision-making process are accessible and interpretable without relying solely on external tools. This attribute is particularly relevant in safety-critical or regulated environments, where traceability and auditability are essential.

- Low Transparency and Explainability: Models whose internal mechanisms are opaque and require external tools or approximations for explanation (also known as black-box models).
- Medium Transparency and Explainability: Models that provide partial visibility into their logic or structure, or that are commonly paired with interpretability frameworks to support explainability.
- High Transparency and Explainability: Models whose internal structure and decision pathways are directly observable and understandable (also known as white-box models).

Transparency complements interpretability by focusing on how decisions are made internally, rather than solely on how easily those decisions can be understood from the outside.

9) MODEL COMPLEXITY (MC)

Model Complexity (MC) captures the architectural depth and computational demands associated with model implementation, training, and deployment. This attribute influences scalability, latency, and suitability for edge or real-time environments.

- Low Complexity: Simple models with shallow architectures, minimal parameterization, and low computational cost.
- Medium Complexity: Moderately complex models with non-linear mappings or ensemble structures, requiring more resources but offering increased flexibility.

- High Complexity: Deep or hybrid architectures with many layers, complex interdependencies, or probabilistic inference mechanisms.

Model complexity must be balanced against infrastructure capacity, transparency needs, and time constraints in industrial prognosis pipelines.

10) ADAPTABILITY TO DATA COMPLEXITY AND GROWTH (ACG)

Adaptability to Data Complexity and Growth (ACG) evaluates a technique's ability to handle complex datasets, as well as its capacity to remain robust as data volume, dimensionality, or structure evolve. This includes managing high-dimensional inputs, nonlinear dependencies, and heterogeneous data sources, whether these characteristics are present from the outset or emerge over time.

- Low Adaptability: Techniques that struggle with high-dimensional or heterogeneous data and may require extensive preprocessing or redesign when the data context changes.
- Medium Adaptability: Techniques that tolerate moderate growth in data complexity and may include mechanisms for feature selection or incremental learning.
- High Adaptability: Techniques that are inherently flexible, scalable, and generalize well under evolving or complex data environments.

High adaptability reduces lifecycle maintenance costs and enhances model longevity in dynamic industrial settings.

11) ALGORITHMIC APPROACH (AA)

Algorithmic Approach (AA) identifies the core methodological family underpinning the technique, such as neural networks, Bayesian models, decision trees, clustering algorithms, or dimensionality reduction methods. This classification provides insight into the technique's theoretical assumptions, training dynamics, typical data representations, and compatibility with available computational infrastructure.

12) APPLICATION DOMAIN (AD)

Indicates the real-world sectors in which the technique has been successfully applied, such as manufacturing, energy, transportation, or healthcare. Although not directly used in the decision pipeline, this attribute offers contextual guidance by indicating whether a technique has been validated in environments with similar operational constraints, regulatory demands, or data conditions. When multiple techniques yield comparable scores, AD can support tie-breaking by highlighting prior use, domain-specific suitability, or maturity within the target sector.

C. ATTRIBUTE CLASSIFICATION AND CODING

To ensure rigor and comparability, each attribute was assigned a classification type, summarized in Table 3:

- Categorical attributes (e.g., LR, DT, AA) are encoded using symbolic codes representing discrete values.
- Ordinal attributes (e.g., IN, TE, MC) use ranked numeric scales for comparative evaluation.

The forty machine learning techniques identified in the survey were classified according to these twelve attributes. This multidimensional characterization complements the technique-stage associations in Table 18, forming the empirical foundation for the filtering and ranking processes presented in the next sections. The attribute structure, proposed in this work, reflects an original synthesis of survey insights and serves as a central contribution of the framework.

TABLE 3. Classification types and coding schemes for machine learning technique attributes.

Attribute	Type of Classification	Possible Values and Coding
Label Requirement (LR)	Categorical	Nr = Not Required Pr = Partially Required Fr = Fully Required
Data Type (DT)	Categorical	St = Structured Sq = Sequential Un = Unstructured Hy = Hybrid
Data Availability (DA)	Categorical	Sc = Data-Scarce Tolerant Mr = Moderate Data Requirement Rr = Data-Rich Required
Automation Level (AL)	Categorical	Mn = Manual Sa = Semi-Automated Fa = Fully Automated
Transferability (TR)	Categorical	Lm = Limited Ad = Adaptable Ht = Highly Transferable
Generality vs. Specialization (GS)	Categorical	Sp = Specialized Md = Moderate Gp = General-Purpose
Interpretability (IN)	Ordinal	1 = Low 2 = Medium 3 = High
Transparency and Explainability (TE)	Ordinal	1 = Low 2 = Medium 3 = High
Model Complexity (MC)	Ordinal	1 = Low 2 = Medium 3 = High
Adaptability to Data Complexity and Growth (ACG)	Ordinal	1 = Low 2 = Medium 3 = High
Algorithmic Approach (AA)	Categorical	Neural Networks; Bayesian; Clustering; Decision Trees; Dimensionality Reduction; etc.
Application Domain (AD)	Categorical	Manufacturing; Healthcare; Transportation; Energy; etc.

Although attribute assignment is inherently interpretative, it followed explicit coding rules to minimize arbitrariness and ensure replicability. Each technique was classified by systematically analyzing the literature evidence and the structured synthesis provided in Table 18, and by applying consistent heuristics:

- Reliance on supervised vs. unsupervised settings determined the LR;

- Recurring data modalities in reported applications defined the DT and informed DA;
- The extent of manual feature engineering and hyperparameter tuning guided the AL;
- Documented cross-domain reuse or domain adaptation approaches supported TR and AD;
- The degree to which techniques were described as broadly applicable versus tailored to specific domains guided the GS;
- Algorithmic families and implementation patterns informed the AA and MC;
- The presence of interpretable decision logic or widely used explainability techniques guided IN, TE, and ACG.

By applying these rules uniformly across the surveyed techniques, the mapping aimed to reduce subjectivity while preserving the literature-grounded character of the taxonomy.

The full mapping is available in Appendix (Table 19), based on evidence from the reviewed literature.

D. FUNCTIONAL CLASSIFICATION OF ATTRIBUTES

The twelve attributes defined in the previous subsection play distinct roles within the proposed framework for selecting ML techniques for fault prognosis. To enable a logical and structured selection process, this work proposes organizing the twelve attributes into three functional categories:

1) Primary Requirements, which comprise six non-compensatory feasibility constraints (LR, DT, DA, AL, TR, and GS) used to exclude incompatible techniques. They are considered indispensable prerequisites to ensure basic compatibility between techniques and application contexts. Among these, LR, DT, and DA are related to data constraints, while AL, TR, and GS reflect application demands. Techniques failing any of these conditions are removed from further consideration, since these attributes capture fundamental feasibility aspects that cannot be offset by strengths in other dimensions.
2) Secondary Evaluation Criteria, which comprise four ordinal attributes (IN, TE, MC, and ACG) used to rank feasible techniques. These attributes were selected since they are universally applicable across ML families, ordinally scalable, and directly capture key industrial deployment priorities: trust and auditability (IN, TE), implementation feasibility (MC), and scalability to evolving data (ACG).
3) Contextual Attributes, which comprise two supporting dimensions (AA and AD) that provide additional insights for interpretation and benchmarking, without directly influencing filtering or ranking. They capture conceptual families and historical use cases, supporting the interpretation of results and assisting in resolving ties when techniques exhibit similar scores.

This functional classification enables the two-phase decision logic of the framework: first, techniques are filtered based on essential feasibility constraints; then, the remaining candidates are ranked to identify the most appropriate option for the application at hand. Table 4 consolidates this classification for quick reference.

With the selection attributes and their functions established, the next subsection outlines the proposed decision pipeline, comprising filtering based on primary requirements and ranking through multi-criteria evaluation.

TABLE 4. Functional classification of selection attributes for ML techniques in fault prognosis.

Attribute	Functional Role
Label Requirement (LR)	Primary Requirement (Data)
Data Type (DT)	Primary Requirement (Data)
Data Availability (DA)	Primary Requirement (Data)
Automation Level (AL)	Primary Requirement (Application)
Transferability (TR)	Primary Requirement (Application)
Generality vs. Specialization (GS)	Primary Requirement (Application)
Interpretability (IN)	Secondary Evaluation Criterion
Transparency and Explainability (TE)	Secondary Evaluation Criterion
Model Complexity (MC)	Secondary Evaluation Criterion
Adaptability to Data Complexity Growth (ACG)	Secondary Evaluation Criterion
Algorithmic Approach (AA)	Contextual Attribute
Application Domain (AD)	Contextual Attribute

E. DECISION REASONING AND APPLICATION FRAMEWORK

The selection process defined in the framework follows a structured, two-phase reasoning pipeline designed to ensure transparent and justifiable decision-making. As shown in Figure 4, modeled using the Business Process Model and Notation (BPMN) formalism, the procedure unfolds in two sequential phases: first, filtering out techniques that do not meet the user's data and application requirements; second, ranking the remaining candidates through a multi-criteria evaluation based on user-defined priorities.

The first step ensures feasibility by eliminating incompatible techniques, while the second supports prioritization through an MCDM method that incorporates user-defined weights and utility functions to reflect context-specific preferences. Although these two steps form the core of the framework, an optional post-selection feasibility assessment is recommended to verify the practical viability of the top-ranked technique before deployment.

The following presents the logic and procedures of each phase, as well as the rationale for the optional assessment step.

1) PHASE 1: PRIMARY REQUIREMENTS FILTERING

The process begins with the user selecting the target prognosis stage: HI construction, HS division, or DTP. This choice defines an initial subset of candidate techniques based on their documented compatibility with the selected stage, as indicated in the Application Context Table (Table 18).

A second filtering step is then applied using the six primary Írequirement attributes defined in Table 4 and detailed in

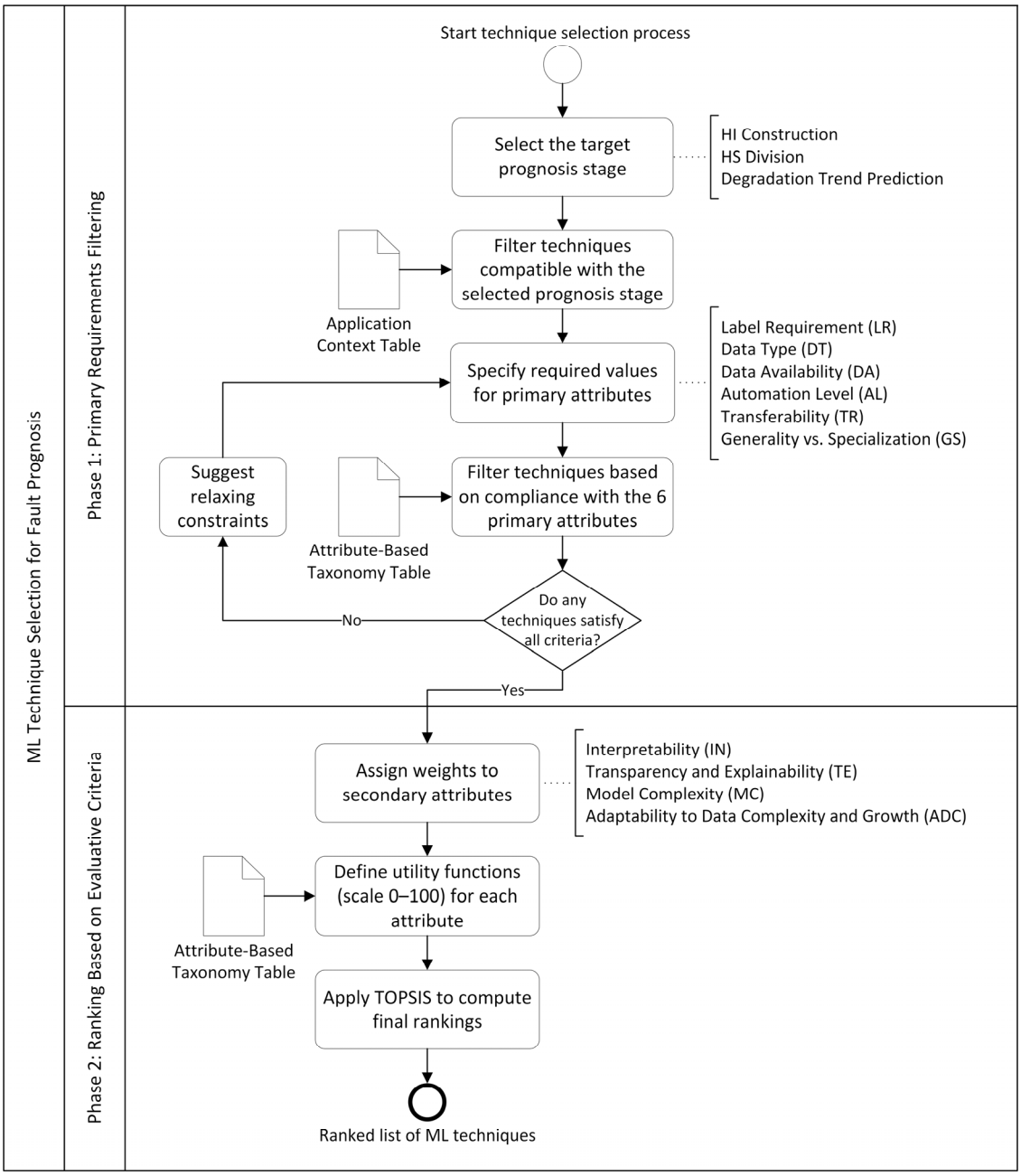


FIGURE 4. Decision-making framework for the selection of ML techniques in fault prognosis.

the Attribute-Based Taxonomy Table (Table 19), with coding conventions specified in

Table 3. These attributes represent essential constraints related to data characteristics and application needs.

The user must specify acceptable values for each of the six criteria. Techniques that do not satisfy all the defined constraints are excluded from further consideration.

2) PHASE 2: RANKING BASED ON EVALUATIVE CRITERIA

Once a compatible subset of techniques has been identified, the second phase ranks the remaining options using a multi-criteria evaluation based on four ordinal attributes from Table 19: IN, TE, MC, and ACG.

The framework employs the Technique for Order Preference by Similarity to Ideal Solution (TOPSIS) [32], [33], which ranks alternatives based on their proximity to an ideal solution and distance from an anti-ideal solution. TOPSIS was selected for its conceptual simplicity, computational efficiency, and established use in engineering contexts. It supports transparent, user-guided prioritization without requiring pairwise comparisons or complex preference functions. Its applicability in scenarios where a single stakeholder guides

model selection aligns with the framework's goals of methodological rigor and practical usability [34], [35].

Considering $A = \{a_1, a_2, \ldots, a_m\}$ the set of m candidate techniques, and $C = \{c_1, c_2, c_3, c_4\}$ the set of four evaluation criteria (IN, TE, MC, ACG), the ranking procedure follows these steps:

1. The user defines a weight vector $\boldsymbol{w} = [w_1, w_2, w_3, w_4]$, where $w_j \in [0, 1]$ and $\sum_{j=1}^{4} w_j = 1$. For each criterion c_j, a discrete utility function $u_j(k)$ is also defined, mapping ordinal levels $k \in \{1, 2, 3\}$ to normalized utility scores in [0, 100], as shown in (1), where, k represents the ordinal score assigned to a technique under a given criterion (1 = "Low", 2 = "Medium", 3 = "High").

$$u_j(k) = \begin{cases} u_j(1), & \text{if } k = 1 \\ u_j(2), & \text{if } k = 2 \\ u_j(3), & \text{if } k = 3 \end{cases} \tag{1}$$

   Utility values can be linear or non-linear, depending on preference sensitivity, such as:
   - $u_j = [0, 50, 100]$: simple linear utility;
   - $u_j = [0, 20, 100]$: convex utility (strong preference for "High");
   - $u_j = [0, 80, 100]$: concave utility (high satisfaction already with "Medium").
2. Utility Matrix Construction: An initial decision matrix $\mathbf{X} = [x_{ij}]$, with $x_{ij} \in \{1, 2, 3\}$, is constructed from Table 19, representing the ordinal score of technique a_i under criterion c_j. Applying the utility functions $u_j(k)$ to each score yields the utility matrix $\mathbf{V} = [v_{ij}]$, with $v_{ij} = u_j(x_{ij})$ being the corresponding utility value obtained.
3. Normalization and Weighted Scoring: The utility matrix $\mathbf{V}$is column-wise normalized to obtain the relative performance matrix $\mathbf{R} = [r_{ij}]$, as in (2).

$$r_{ij} = \frac{v_{ij}}{\sqrt{\sum_{i=1}^{m} v_{ij}^2}} \tag{2}$$

   Next, multiplying $\mathbf{R}$ by the weight vector produces the weighted matrix $\mathbf{Z} = [z_{ij}]$, where z_{ij} is given by (3).

$$z_{ij} = w_j r_{ij} \tag{3}$$

4. Ideal and Anti-Ideal Solutions Identification: From $\mathbf{Z}$, the best and worst scores across all criteria define the ideal and anti-ideal solutions, as shown in (4) and (5), respectively.

$$\mathbf{z}^+ = \left\{ z_j^+ = \max_i \left(z_{ij}\right) \mid j = 1, 2, 3, 4 \right\} \tag{4}$$

$$\mathbf{z}^- = \left\{ z_j^- = \min_i \left(z_{ij}\right) \mid j = 1, 2, 3, 4 \right\} \tag{5}$$

5. Distance Calculation: For each candidate technique a_i, the Euclidean distances to the ideal and anti-ideal solutions are computed using (6) and (7), respectively.

$$D_i^+ = \sqrt{\sum_{j=1}^{4} \left(z_{ij} - z_j^+\right)^2} \tag{6}$$

$$D_i^- = \sqrt{\sum_{j=1}^{4} \left(z_{ij} - z_j^-\right)^2} \tag{7}$$

6. TOPSIS Score Computation: Each technique's relative closeness to the ideal is given by its TOPSIS score C_i, defined in (8). Values closer to 1 indicate better overall performance.

$$C_i = \frac{D_i^-}{D_i^+ + D_i^-} \tag{8}$$

7. Final Ranking: Techniques are ranked in descending order of C_i. The highest-scoring option is considered the most suitable, given the user's defined priorities and constraints.

3) POST-SELECTION FEASIBILITY ASSESSMENT

Although the two-phase procedure provides a structured method for identifying and ranking candidate techniques for fault prognosis, practical deployment may still face unforeseen constraints. The top-ranked technique, although optimal in terms of the evaluated criteria, may prove infeasible within a specific operational or organizational context. To address this, a post-selection feasibility assessment is recommended as a final validation step before implementation.

This assessment consists of a contextual analysis of the top-ranked technique, and, if needed, its immediate alternatives, across four complementary dimensions:

- Operational Feasibility: Verifies whether the technique can be integrated into the current infrastructure, considering hardware requirements (e.g., memory, real-time inference), software compatibility, and data acquisition pipelines.
- Organizational Readiness: Assesses whether the necessary technical expertise exists to develop, interpret, and maintain the model. Where gaps are identified, the cost and timeline for capacity-building or external support should be considered.
- Non-Technical Constraints: Includes regulatory, explainability, cybersecurity, and integration concerns. Techniques lacking transparency, for example, may not be acceptable in regulated settings.
- Alternative Trade-Offs: Encourages a comparative review of the top three ranked techniques, weighing performance scores against implementation risk and long-term viability. In some cases, a slightly lower-ranked method may offer a more balanced and sustainable solution.

By incorporating this final layer, the framework reinforces its practical relevance and mitigates the risk of selecting contextually unsuitable solutions. This step strengthens decision-making robustness, particularly in industrial environments with complex and dynamic constraints.

V. APPLICATION EXAMPLE AND DISCUSSION

This section presents a real-world application example that illustrates the step-by-step use of the proposed parameterized

TABLE 5. Mapping of primary attribute constraints guiding phase 1 of the framework.

Attribute	Target Value	Code	Justification
Label Requirement (LR)	Not Required or Partial Required	Nr/Pr	Failure events are not labeled; only unsupervised or self-supervised techniques are applicable.
Data Type (DT)	Structured or Sequential	St/Sq	Sensor data are stored as time-stamped SCADA logs and multivariate time series.
Data Availability (DA)	Data-Scarce Tolerant to Moderate	Sc/Mr	Only recent operational data are available; long-term data histories are missing.
Automation Level (AL)	Semi-Automated or Fully Automated	Sa/Fa	There are no ML specialists available; solutions must require minimal manual intervention.
Transferability (TR)	Not Constrained	-	Each model will be trained per equipment; broad transferability is not required.
Generality vs. Specialization (GS)	Moderate to General-Purpose	Md/Gp	The technique must support multiple classes of equipment without relying on asset-specific tuning.

TABLE 6. Mapping of secondary attribute constraints used in phase 2 of the framework.

Attribute	Weight	Utility	Attribute
Interpretability (IN)	0.40	[0, 60, 100]	Interpretability ensures that outputs, such as HI or failure probabilities, are clear and trustworthy for operators without ML expertise. The moderately convex utility captures the increasing value of models aligned with domain understanding.
Transparency & Explainability (TE)	0.20	[0, 80, 100]	Transparency supports traceability and confidence in automated decisions. The convex utility favors techniques that reveal internal logic or key features, especially under limited supervision.
Model Complexity (MC)	0.20	[100, 20, 0]	In automated settings with minimal expert input, simpler models are easier to deploy, maintain, and scale. The strongly inverse utility penalizes high-complexity techniques with costly architectures or processing demands.
Adaptability to Data Complexity & Growth (ACG)	0.20	[0, 20, 100]	As sensors and data complexity increase, models must remain robust and scalable. The sharply increasing utility favors techniques that tolerate growth with minimal reengineering.

framework for selecting machine learning techniques in fault prognosis. The case addresses a practical scenario in the energy sector encountered by the authors: a recently acquired electrical power substation is being integrated into a predictive maintenance routine, yet no labeled failure data are available due to the absence of historical records.

The substation is equipped with structured, time-stamped sensor data collected from key assets such as power transformers and circuit breakers. The number of monitored variables is expected to increase as additional components are instrumented. The main objective is to estimate the probability of failure within a predefined time horizon (e.g., 30 days) for each asset, supporting maintenance planning and operational risk mitigation.

The selected solution must be scalable across equipment types and usable by maintenance engineers without requiring dedicated machine learning specialists. No assumptions are made regarding which technique performs best. Instead, the example follows the framework's two-phase decision process, from defining operational constraints to ranking suitable alternatives, offering a transparent and reproducible approach to technique selection under real-world energy infrastructure conditions.

A. CONSTRAINTS AND REQUIREMENTS

Given the nature of the available information, this application involves two stages of the prognosis process: HI construction and DTP. Since data are directly obtained from SCADA systems and monitoring logs, a separate data acquisition phase is not required.

The goal is to estimate the FPPH based on current sensor behavior. As the system operates under stable conditions and degradation is expected to occur gradually, the data do not exhibit clear transitions between health stages.

Moreover, there is insufficient information to model such transitions reliably. Therefore, the risk model must capture degradation as a continuous trend through the HI, without segmenting it into discrete health regimes.

The primary constraints guiding Phase 1 of the framework are summarized in Table 5. In Phase 2, the four evaluative attributes are used to rank viable techniques, as shown in Table 6. A reverse utility function is applied to MC to penalize overly complex models, while the remaining criteria follow a direct (increasing) utility orientation.

The same constraints (Table 5 and Table 6) apply to both HI construction and DTP stages. This alignment reflects a shared operational context, including infrastructure, data structure, and deployment conditions. Since both models are developed and maintained under the same constraints, selection criteria and decision priorities remain consistent across stages.

B. STEP-BY-STEP USE OF THE FRAMEWORK

The framework is applied in two stages of the prognosis process: HI construction and DTP. For each, the procedure begins

by identifying compatible techniques, followed by filtering based on primary constraints and ranking via secondary criteria.

For HI construction, 18 techniques are initially considered, covering statistical methods (e.g., ARMA, PCA), signal processing approaches (e.g., Fourier and Wavelet transforms), and machine learning models (e.g., Autoencoder, SOM). For DTP, the taxonomy includes 31 techniques capable of modeling temporal degradation dynamics, ranging from traditional time-series models (e.g., ARIMA, Kalman Filter) to advanced machine learning and ensemble methods (e.g., Clustering, Transfer Learning).

Due to the absence of labeled failure events, only techniques supporting unsupervised (i.e., no label usage) or self-supervised learning (i.e., label-free training based on surrogate tasks such as reconstruction or contrastive prediction) are admissible. This reduces the candidate set to nine techniques for HI construction (ARMA, Autoencoder, Fourier Transform, Fuzzy Logic, GMM, Kalman Filter, PCA, SOM, and Wavelet) and eight for DTP (ARIMA, Brownian Motion, Clustering, Fuzzy Logic, Kalman Filter, KBES, Transfer Learning, and Wavelet). All selected techniques are compatible with structured or sequential data and tolerate data-scarce conditions, consistent with SCADA logs and the limited operational history of the substation. No techniques are excluded at this stage.

Next, techniques requiring extensive manual configuration are removed. For HI construction, the Fourier Transform and Fuzzy Logic are excluded due to their reliance on manually defined parameters or rule sets. The remaining techniques are: ARMA, Autoencoder, GMM, Kalman Filter, PCA, SOM, and Wavelet.

For DTP, Brownian Motion, Fuzzy Logic, and KBES are excluded for similar reasons, leaving: ARIMA, Clustering, Kalman Filter, Transfer Learning, and Wavelet. Among these, only Clustering is fully automated (Fa), while the others are semi-automated (Sa).

As models are trained independently per asset, the transferability constraint is not enforced. Regarding Generality vs. Specialization, most retained techniques are classified as moderately general (Md), with PCA and Transfer Learning categorized as general-purpose (Gp) due to their broad applicability.

At the end of Phase 1, seven techniques remain for HI construction and five for DTP. These are ranked using the TOPSIS method, applying the attribute weights and utility functions defined in Table 6. Because both stages operate in the same environment, the same evaluation configuration is used.

Tables 7 and 8 present the ordinal evaluations of the shortlisted techniques, extracted from Table 19.

Using the defined utility functions, these values are converted into weighted matrices, shown in Tables 9 and 10. In turn, the ideal and anti-ideal solutions are identified by selecting the highest and lowest values for each attribute, respectively (Tables 11 and 12). These vectors represent the best and worst performance profiles according to the user-defined priorities.

TABLE 7. Secondary attribute values for hi construction techniques.

Technique Name	IN	TE	MC	AGC
ARMA	3	3	1	1
Autoencoder	2	1	2	3
GMM	2	2	2	2
Kalman Filter	3	3	2	2
PCA	3	3	1	2
SOM	2	2	2	2
Wavelet	2	2	1	2

TABLE 8. Secondary attribute values for DTP techniques.

Technique Name	IN	TE	MC	AGC
ARIMA	3	3	1	1
Clustering	2	2	2	2
Kalman Filter	3	3	2	2
Transfer Learning	2	2	2	3
Wavelet	2	2	1	2

TABLE 9. Weighted matrix for HI construction techniques.

Technique Name	IN	TE	MC	AGC
ARMA	0.1898	0.0902	0.1125	0.0000
Autoencoder	0.1139	0.0000	0.0225	0.1826
GMM	0.1139	0.0721	0.0225	0.0365
Kalman Filter	0.1898	0.0902	0.0225	0.0365
PCA	0.1898	0.0902	0.1125	0.0365
SOM	0.1139	0.0721	0.0225	0.0365
Wavelet	0.1139	0.0721	0.1125	0.0365

TABLE 10. Weighted matrix for DTP techniques.

Technique Name	IN	TE	MC	AGC
ARIMA	0.2279	0.1010	0.1374	0.0000
Clustering	0.1368	0.0808	0.0275	0.0378
Kalman Filter	0.2279	0.1010	0.0275	0.0378
Transfer Learning	0.1368	0.0808	0.0275	0.1890
Wavelet	0.1368	0.0808	0.1374	0.0378

TABLE 11. Ideal and anti-ideal solutions for HI construction.

Solution	IN	TE	MC	AGC
Ideal	0.1898	0.0902	0.1125	0.1826
Anti-Ideal	0.1139	0.0000	0.0225	0.0000

TABLE 12. Ideal and anti-ideal solutions for DTP.

Solution	IN	TE	MC	AGC
Ideal	0.2279	0.1010	0.1374	0.1890
Anti-Ideal	0.1368	0.0808	0.0275	0.0000

Each candidate technique is evaluated based on its Euclidean distance to both the ideal and anti-ideal solutions. The TOPSIS score C is then computed as the ratio between

the distance to the anti-ideal and the total distance to both reference points. The final scores and rankings for each stage are presented in Tables 13 and 14.

TABLE 13. TOPSIS scores for HI construction techniques.

Technique Name	D^+	D^-	C
Autoencoder	0.021997	0.033333	0.602441
PCA	0.021333	0.023330	0.522357
ARMA	0.033333	0.021997	0.397559
Wavelet	0.027424	0.014638	0.348005
Kalman Filter	0.029435	0.015229	0.340974
GMM	0.035526	0.006537	0.155403
SOM	0.035526	0.006537	0.155403

TABLE 14. TOPSIS scores for DTP techniques.

Technique Name	D^+	D^-	C
Transfer Learning	0.020795	0.035714	0.632004
ARIMA	0.035714	0.020795	0.367996
Wavelet	0.031577	0.013504	0.299550
Kalman Filter	0.034933	0.010148	0.225115
Clustering	0.043652	0.001429	0.031689

The results indicate that the Autoencoder obtained the highest TOPSIS scores for HI construction. Autoencoder was favored for its ability to extract meaningful indicators in an unsupervised manner and for its adaptability to growing data complexity. PCA and ARMA followed closely, offering high interpretability and simplicity, albeit with more limited adaptability.

For DTP, Transfer Learning stood out for its flexibility and potential for generalization in evolving data environments. However, a Post-Selection Feasibility Assessment revealed that Transfer Learning was not suitable for this application. Its effective use would require pretrained models or related labeled datasets, which are not available. Moreover, its deployment would involve complex architectures and fine-tuning procedures incompatible with the system's automation requirements and limited technical supervision. For these reasons, ARIMA, the second-ranked technique, was selected instead. It offers low complexity, high interpretability, and ease of integration – characteristics that better align with the project's operational constraints.

This adjustment underscores the value of complementing quantitative evaluation with contextual feasibility checks, ensuring that selected techniques are not only optimal on paper but also viable in practice.

C. RESULTS DISCUSSION

The framework's application revealed not only the final recommendations for each prognosis stage, but also the rationale behind them, i.e., how data constraints, evaluation priorities, and contextual requirements interact to shape the selection of machine learning techniques. In this case, Autoencoder (for HI construction) and ARIMA (for DTP, following feasibility reassessment) emerged as the most suitable techniques given the absence of labeled failure data, the need for automation, and the importance of interpretability to support maintenance planning.

The role of the framework is not to produce absolute answers, but to guide practitioners through a transparent and consistent decision-making process. Final choices must be interpreted critically, especially when techniques show close scores or share conceptual foundations. For example, in HI construction, Autoencoder and PCA achieved similar TOPSIS scores (0.602 and 0.522), and both rely on unsupervised dimensionality reduction to generate compact indicators from multivariate data. In such cases, the selection can be refined by incorporating additional validation experiments or contextual considerations, such as Algorithmic Approach or Application Domain, to support the final deployment.

The same principle applies to feasibility. Although Transfer Learning ranked highest for DTP, it was replaced by ARIMA due to operational constraints, including the lack of pretrained models and the need for low-complexity deployment. This reinforces a key aspect of the framework: results must be reviewed through technical, operational, and organizational lenses before implementation.

A similar situation arises with GMM and SOM in the HI construction stage, which received identical TOPSIS scores in the final ranking position. While this specific tie may not affect the present decision, similar cases could be more consequential in practice. In such instances, contextual attributes like AD may aid in tie-breaking by highlighting historical precedent, domain-specific relevance, or implementation maturity. For example, although both GMM and SOM have been applied in manufacturing (Table 19), SOM has a longer track record in fault prognosis (as shown in Figure 3), potentially indicating greater maturity or broader availability of domain-specific expertise.

To test the framework's robustness and adaptability, a sensitivity analysis was conducted under three scenarios. The first two focus on controlled variations in key input assumptions: (i) assuming the availability of fully labeled failure data (with other constraints unchanged) and (ii) adopting simplified evaluation preferences, using uniform weights and basic utility functions. The third scenario addresses ranking consistency, assessing how frequently each technique occupied specific positions across multiple simulations and how stable these positions remained under changes in evaluation priorities. Together, these analyses provide a broader view of the stability, interpretability, and operational relevance of the framework's recommendations.

1) SENSITIVITY TO LABELING CONSTRAINTS

In the first sensitivity scenario, the Label Requirement constraint was modified from Not Required (Nr) or Partially Required (Pr) to Fully Required (Fr), simulating the availability of labeled failure data for all monitored assets. This single change substantially altered the set of admissible techniques for both prognosis stages.

For HI construction, only two techniques passed the updated filter: Random Forest, which achieved a TOPSIS score of 1.000, and Time Series Analysis, with a score of 0.000. This stark contrast illustrates how restrictive constraints can sharply narrow the solution space, often highlighting a single dominant option.

For DTP, the change yielded a broader candidate set, with eleven techniques passing the filters. The top three were Random Forest ($C = 0.674$), Decision Tree ($C = 0.629$), and Stacking ($C = 0.605$). These techniques share structural similarities: they are supervised, tree-based, and suited to non-linear or imbalanced problems. In addition, they are conceptually linked through ensemble logic. Random Forest can be viewed as a special case of Stacking, in which multiple decision trees are combined via bagging with random feature selection. Stacking generalizes this idea by combining diverse learners into a meta-model. The prominence of this trio reflects the framework's internal coherence, consistently pointing to ensemble decision-tree models as the most suitable under fully labeled conditions.

Notably, Random Forest ranked first in both stages, suggesting that a single technique could address the entire prognosis process. This has important operational implications: it simplifies deployment, reduces the learning curve for maintenance teams, and may support unified diagnostic-prognostic pipelines.

However, it is essential to interpret high-ranking options in context. Some techniques, like Stacking, are better understood as modeling strategies rather than stand-alone solutions. Stacking would depend on the definition and availability of base learners and a meta-learner to function effectively. The same applies to techniques like Transfer Learning, Wavelet, or Fourier Analysis, which often serve as components within broader pipelines, supporting tasks such as signal decomposition, feature extraction, or model reuse, rather than directly delivering predictions.

Recognizing these distinctions is critical for interpreting the framework's outputs. Ranking scores must be contextualized with an understanding of each technique's scope, purpose, and integration potential. Overall, this scenario confirms that the framework adapts to variations in data availability while supporting the identification of coherent, structurally aligned model families. It reinforces the need for practitioner judgment in validating results based on algorithmic design, domain logic, and implementation feasibility.

2) SENSITIVITY TO EVALUATION PREFERENCES

In the second sensitivity scenario, the configuration of secondary attributes was modified to assess how changes in evaluation priorities might influence the outcome. The original weight vector (0.40, 0.20, 0.20, 0.20) and non-linear utility functions were replaced with uniform weights (0.25 for all attributes) and simple linear utilities: [0, 50, 100] for IN, TE, and AGC, and [100, 50, 0] for MC. This configuration simulates a context in which all attributes are equally important, and preferences grow linearly with attribute levels (or decrease linearly for model complexity).

Despite this simplification, the framework continued to yield coherent and interpretable rankings. Under the original data constraints (no labeled failures), PCA ranked first for HI construction ($C = 0.819$), followed by Kalman Filter and Autoencoder. For DTP, Transfer Learning remained at the top ($C = 0.677$), with Kalman Filter and Wavelet next. Once again, Transfer Learning was excluded in the final choice due to feasibility, leading to the selection of the Kalman Filter. Combined with PCA, this pairing forms a technically consistent pipeline: PCA constructs compact health indicators, and the Kalman Filter estimates the evolution of the latent degradation state over time.



FIGURE 5. Heatmap of ranking frequencies per HI construction technique.

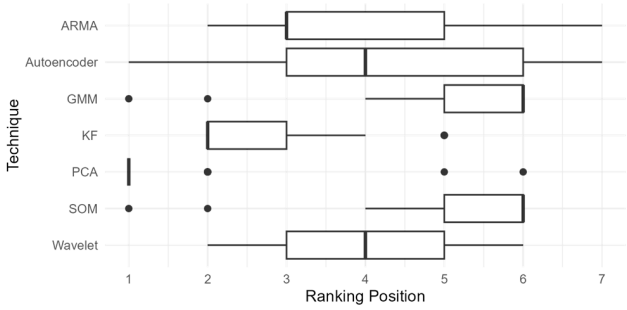


FIGURE 6. Boxplot of position distributions per HI construction technique.

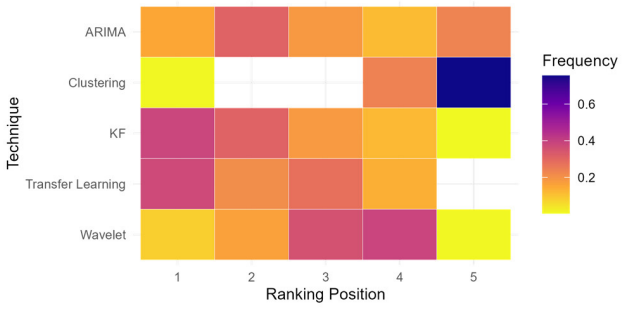


FIGURE 7. Heatmap of ranking frequencies per DTP technique.

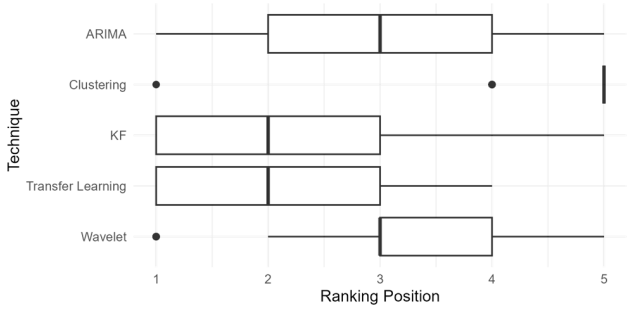


FIGURE 8. Boxplot of position distributions per DTP technique.

When the same uniform configuration was applied to the fully labeled data scenario, the results shifted as expected. Random Forest remained the top choice for HI construction ($C = 1.000$), while Time Series Analysis remained last. In DTP, Decision Tree emerged as the new leader ($C = 0.871$), followed by Random Forest, Linear Regression, Logistic Regression, and Stacking. This group reflects internal coherence, as all are widely used in supervised prognosis and are consistent with the availability of labeled data.

Importantly, even when weights and utility functions are simplified, the framework continues to prioritize techniques compatible with the informational and operational profile of each case. This reinforces its robustness: results remain stable and meaningful even when preference structures are loosely defined.

Some variation in technique ordering is expected and, in fact, desirable. The purpose of multi-criteria selection is not to prescribe fixed outcomes, but to support the structured exploration of trade-offs. The emergence of different techniques under varying evaluative schemes encourages practitioners to reflect on what matters most in their context and to justify their final choice accordingly.

In this sense, the framework remains both prescriptive and dialogical: it structures decisions while preserving the practitioner's agency and critical role in the final selection.

To further assess the robustness of the weighting scheme and preference structures, the next subsection evaluates ranking stability across a wide range of weight and utility variations.

3) RANKINGS CONSISTENCY ANALYSIS

In the third sensitivity scenario, the empirical stability of the selection outcomes was assessed through a consistency analysis based on a broad set of simulated ranking scenarios for each prognosis stage. These simulations reflect variations in evaluation priorities, including both the weight vector assigned to each criterion and the shape of the utility functions, particularly the central reference point that determines whether preferences grow linearly, concavely, or convexly with attribute levels. Throughout this analysis, the Label Requirement constraint was kept as originally defined, Not Required (Nr), ensuring that the candidate set of techniques remained identical to that obtained in the Primary Requirements Filtering phase.

While each ranking position derives from the TOPSIS score obtained in each simulation, this analysis shifts the focus from individual scores to the distribution of positions across multiple runs. By examining how often each technique occupied each rank, and how concentrated or dispersed its positional behavior was, it becomes possible to assess the empirical consistency of the framework's outputs under varying preference configurations.

The simulations explored a wide range of evaluation contexts by maximizing variability in attribute weights and utility shapes. A total of 265 runs were generated. Although not all possible ranking permutations were simulated, analysis of cumulative medians and bootstrap-based uncertainty bands showed that position estimates stabilized after about 200 runs. This convergence indicates that the sample was sufficient to capture the underlying variability in technique rankings, supporting the reliability of the frequency-based consistency analysis.

Two visualizations were used to assess ranking consistency across simulations. Heatmaps show how often each technique occupied each position, revealing dominance when frequencies concentrate in the upper-left corner and sensitivity to preferences when positions are more dispersed. Boxplots complement this view by summarizing the full position distribution for each technique, highlighting the median, interquartile range, and outliers. Together, they provide a concise picture of central tendency, variability, and overall ranking stability.

Figures 5 and 6 present the heatmap and boxplot, respectively, for the techniques selected for HI construction, while Figures 7 and 8 show the corresponding results for DTP.

The analysis for HI construction (Figures 5 and 6) reveals a clear dominance of PCA, which consistently appears in the first position in over 80% of the simulations. This is further confirmed by its boxplot, which shows a median of 1 with minimal dispersion. In contrast, Autoencoder spans almost the entire ranking scale (positions 1 to 7), with a median at position 4, reflecting both its potential and sensitivity to variations in user-defined preferences.

Wavelet, also with a median at 4, shows a more concentrated range between positions 3 and 5, but never reaches the top (position 1) or the bottom (position 7), suggesting moderate but limited competitiveness. KF also stands out as a consistent option, with a median at position 2. While it does not outperform PCA, its declining probability toward lower ranks indicates stable performance within a narrow band.

ARMA exhibits a similar pattern, albeit more diversified: it has a median at position 3, an interquartile range from 3 to 5, and a non-zero probability of achieving position 2, while also being occasionally ranked among the lowest. Techniques such as GMM and SOM display virtually identical behaviors. Although they never occupy the last position, they remain confined to the lower half of the ranking spectrum, with a

TABLE 15. Glossary of machine learning and signal processing techniques referenced in the fault prognosis literature.

Index	Technique Name	Index	Technique Name
1	Adaptive Boosting	90	Matrix Factorization
2	Adaptive Resonance Theory	91	Memory Networks
3	Anomaly Detection	92	Meta-Learning
4	Attention Mechanisms Beyond Transformers	93	Metric Learning
5	Auto Regressive Integrated Moving Average	94	Model Compression
6	Auto Regressive Moving Average	95	Model Pruning
7	Autoencoder	96	Model-Based Reinforcement Learning
8	Automated Machine Learning	97	Monte Carlo Tree Search
9	Bayesian Networks	98	Multimodal Learning
10	Bayesian techniques with Kalman Filters	99	Multimodal Retrieval
11	Bayesian techniques with Particle Filters	100	Multinomial Logistic Regression
12	Boosted Tree	101	Multiple Linear Regression
13	Bootstrap Aggregating	102	Naive Bayes
14	Brownian Motion	103	Named Entity Recognition
15	Capsule Networks	104	Neighborhood Component Analysis
16	Categorical Boosting	105	Neighborhood-based Methods
17	Causal Inference	106	Network Embedding
18	Clustering	107	Neural Architecture Search
19	Collaborative Filtering	108	Neural Ordinary Differential Equations
20	Continuous Learning	109	Neuroevolution
21	Convolutional Neural Network	110	Neuromorphic Computing
22	Convolutional Neural Network based Long Short-Term Memory	111	Non-Negative Matrix Factorization
23	Correspondence Analysis	112	One-Class Support Vector Machine
24	Co-training	113	Ordering Points to Identify Clustering Structure
25	Counterfactual Learning	114	PageRank
26	Cross-Modal Retrieval	115	Particle Filter
27	Decision Tree	116	Part-of-Speech Tagging
28	Deep Autoencoder	117	Principal Component Analysis
29	Deep Belief Network	118	Principal Component Analysis Self Organizing Map
30	Deep Bidirectional Long Short-Term Memory Network	119	Proportional Hazard Models
31	Deep Generative Models	120	Proximal Policy Optimization
32	Deep Learning Architectures	121	Quadratic Programming
33	Deep Neural Network	122	Quantile Regression
34	Deep Reinforcement Learning	123	Quantum Machine Learning
35	Density-Based Spatial Clustering of Applications with Noise	124	Radial Basis Network
36	Dimensionality Reduction Techniques	125	Random Forest
37	Elastic Net	126	Random Walk on Networks
38	Ensemble Learning Variants	127	Random Walks
39	Ensemble of Neural Networks	128	Recurrent Neural Network
40	Evolutionary Algorithms	129	Reinforcement Learning
41	Exponential Smoothing	130	Relevance Vector Machine
42	Extreme Gradient Boosting	131	Replicator Neural Networks
43	Extreme Learning Machines	132	Ridge Regression
44	Few-Shot Learning	133	Rough Sets
45	Fourier Transform	134	Rule-based Systems
46	Fuzzy Logic	135	Self-Organizing Map (Kohonen Map)
47	Gaussian Mixture Model	136	Self-Attention Mechanisms
48	Gaussian Process Classification	137	Semi-Markov Models
49	Gaussian Process Regression	138	Semi-Supervised Support Vector Machine
50	Generative Adversarial Networks	139	Short-Time Fourier Transform
51	Gradient Boosted Regression Trees	140	Siamese Networks
52	Gradient-Boosted Trees	141	Silhouette
53	Graph Attention Networks	142	Simulated Annealing
54	Graph based model	143	Simultaneous Localization and Mapping
55	Graph Convolutional Networks	144	Social Network Analysis
56	Graph Neural Networks	145	Spatial Data Analysis
57	Graph-Based Models	146	Spectral Clustering
58	Hidden Markov Model-Based Speech Recognition	147	Stacked Autoencoder
59	Hidden Markov Models	148	Stacking
60	Hidden Semi-Markov Models	149	State Space Models
61	Hilbert-Huang Transform	150	Static Bayesian Networks
62	Homomorphic Encryption	151	Statistical Pattern Recognition
63	Hybrid Recommender Systems	152	Statistical Trend Extrapolation
64	Imitation Learning	153	Stochastic Aggregation
65	Independent Component Analysis	154	Support Vector Classification
66	Instance-based Learning	155	Support Vector Machine
67	Isolation Forest	156	Support Vector Machines Ensembles
68	Kalman Filter	157	Support Vector Regression
69	Kernel Density Estimation	158	Swarm Intelligence
70	Kernel Methods	159	T-Distributed Stochastic Neighbor Embedding

TABLE 15. *(Continued.)* **Glossary of machine learning and signal processing techniques referenced in the fault prognosis literature.**

71	K-Means	160	Temporal Convolutional Networks
72	K-Nearest Neighbors	161	Temporal Difference Learning
73	Knowledge Based Expert Systems	162	Text Generation
74	Knowledge Distillation	163	Time Domain Analysis
75	Knowledge Graphs	164	Time Series Analysis
76	Ladder Network	165	Time Series Forecasting
77	Lattice Random Walk	166	Topological Data Analysis
78	Learning to Rank	167	Transfer Learning
79	Least Absolute Shrinkage and Selection Operator	168	Transfer Learning with Generative Models
80	Light Gradient Boosting Machine	169	Transfer Learning with Language Models
81	Linear Discriminant Analysis	170	Transformer Models
82	Linear Regression	171	Triplet Networks
83	Local Outlier Factor	172	Variational Autoencoders
84	Logistic Regression	173	Variational Inference
85	Long Short-Term Memory	174	Wavelet
86	Long Short-Term Memory and Gated Recurrent Unit	175	Wavelet Packet Energie Spectrum
87	Long Short-Term Memory Restricted Boltzmann Machine	176	Wigner-Ville Distribution
88	Markov Chain Monte Carlo	177	Word Embeddings
89	Match Matrix	178	Word2Vec

shape that mirrors that of the KF, yet shifted downward in performance.

For DTP (Figures 7 and 8), Clustering demonstrates markedly poor performance, occupying the last position (5) in over 80% of the runs and never appearing in better-ranked positions. On the other hand, Transfer Learning and KF show favorable central tendencies (median = 2), although with considerable dispersion.

Notably, Transfer Learning never ranks last, underscoring its resilience to shifts in evaluation criteria. ARIMA has a median around position 3, with broader variability spanning all positions (1 to 5), reflecting moderate performance that is more sensitive to preference trade-offs. Wavelet, while also centered around position 3, shows a tendency toward lower rankings, confirming a relatively weaker performance.

Overall, this consistency analysis reinforces the robustness of the framework. Techniques with strong conceptual alignment and compatibility with the data constraints tend to retain high positions across a wide range of evaluation scenarios. Conversely, techniques that lack structural suitability or are sensitive to specific trade-offs exhibit more scattered or persistently low rankings. These insights highlight not only the most promising techniques, but also the stability and limitations of each choice under realistic variations in evaluator priorities.

Notably, the strong and recurrent dominance of PCA and, to a lesser extent, Autoencoder for HI construction suggests that, under the given constraints, dimensionality reduction emerges as a central modeling strategy. The ability of these techniques to condense multivariate signals into compact, degradation-sensitive indicators appears particularly well-suited to unsupervised and interpretable settings, reinforcing their relevance in early-stage prognosis tasks when labeled failure data is unavailable.

It is important to emphasize, however, that this analysis considers only the variability induced by changes in evaluation preferences, e.g., weight allocations and utility function shapes, while keeping fixed the candidate set previously defined in Phase 1 (Primary Requirements Filtering). As this initial filtering phase does not rank techniques but rather excludes those that fail to meet core feasibility criteria, the current results reflect the comparative behavior of admissible options only.

VI. CONCLUSION

This study presented a structured framework for selecting machine learning techniques in fault prognosis, supporting decision-making in industrial contexts with diverse data conditions, operational constraints, and implementation priorities. While motivated by a real-world case in the power sector, the framework was conceived as domain-agnostic and adaptable to a wide range of condition-based maintenance and predictive analytics applications.

Grounded in a comprehensive literature survey and organized around a two-phase selection logic, the framework enables users to filter incompatible techniques based on feasibility constraints and to rank viable alternatives through a flexible, user-driven multi-criteria evaluation.

The framework offers several strengths. First, it separates primary constraints, such as label requirements, data format, and automation level, from secondary evaluation criteria, including interpretability, complexity, and adaptability. This layered structure supports a stepwise decision process: narrowing the field and then ranking candidates in line with context-specific goals.

TABLE 16. Frequency of machine learning and signal processing terms in the fault prognosis literature (metadata occurrences).

Technique Name	Frequency	Technique Name	Frequency
Wavelet	295	Bootstrap Aggregating	6
Convolutional Neural Network	213	Deep Learning Architectures	6
Long Short-Term Memory	205	Exponential Smoothing	6
Principal Component Analysis	185	State Space Models	6
Support Vector Machine	185	T-Distributed Stochastic Neighbor Embedding	6
Anomaly Detection	142	Temporal Convolutional Networks	6
Clustering	120	Adaptive Boosting	5
Particle Filter	113	Adaptive Resonance Theory	5
Kalman Filter	110	Continuous Learning	5
Autoencoder	97	Graph Convolutional Networks	5
Transfer Learning	93	Isolation Forest	5
Hidden Markov Models	79	Least Absolute Shrinkage and Selection Operator	5
Support Vector Regression	71	Metric Learning	5
Recurrent Neural Network	67	Multinomial Logistic Regression	5
Fourier Transform	49	Ridge Regression	5
Deep Neural Network	48	Time Domain Analysis	5
Fuzzy Logic	48	Variational Autoencoders	5
Memory Networks	46	Deep Reinforcement Learning	4
Knowledge-Based Expert Systems	43	Local Outlier Factor	4
Random Forest	43	Markov Chain Monte Carlo	4
Bayesian Networks	42	One-Class Support Vector Machine	4
Self-Organizing Map (Kohonen Maps)	41	Simulated Annealing	4
Deep Belief Network	38	Transformer Models	4
Generative Adversarial Networks	38	Evolutionary Algorithms	3
Decision Tree	35	Few-Shot Learning	3
Extreme Learning Machines	32	Kernel Methods	3
Semi-Markov Models	32	Meta-Learning	3
Brownian Motion	31	Quantum Machine Learning	3
Gaussian Process Regression	31	Capsule Networks	2
Linear Regression	28	Causal Inference	2
Relevance Vector Machine	28	Co-training	2
Gaussian Mixture Model	26	Elastic Net	2
Time Series Analysis	26	Matrix Factorization	2
Logistic Regression	24	Model Pruning	2
K-Means	23	Neural Architecture Search	2
K-Nearest Neighbors	23	Non-Negative Matrix Factorization	2
Deep Bidirectional Long Short-Term Memory Network	21	Ordering Points to Identify Clustering Structure	2
Hidden Semi-Markov Models	19	Quadratic Programming	2
Short-Time Fourier Transform	19	Quantile Regression	2
Auto Regressive Moving Average	17	Radial Basis Network	2
Stacked Autoencoder	16	Spectral Clustering	2
Auto Regressive Integrated Moving Average	14	Swarm Intelligence	2
Hilbert-Huang Transform	14	Automated Machine Learning	1
Kernel Density Estimation	14	Categorical Boosting	1
Linear Discriminant Analysis	14	Deep Generative Models	1
Stacking	14	Dimensionality Reduction Techniques	1
Reinforcement Learning	13	Gaussian Process Classification	1
Time Series Forecasting	13	Gradient-Boosted Trees	1
Independent Component Analysis	12	Graph-Based Models	1
Multiple Linear Regression	11	Instance-based Learning	1
Deep Autoencoder	10	Long Short-Term Memory and Gated Recurrent Unit	1
Naive Bayes	9	Named Entity Recognition	1
Statistical Pattern Recognition	9	Neighborhood Component Analysis	1
Self-Attention Mechanisms	8	Proportional Hazard Models	1
Static Bayesian Networks (Directed Acyclic Graphs)	8	Random Walks	1
Bayesian techniques with Particle Filters	7	Rough Sets	1
Density-Based Spatial Clustering of Applications with Noise	7	Social Network Analysis	1
Extreme Gradient Boosting	7	Support Vector Classification	1
Graph Neural Networks	7	Support Vector Machines Ensembles	1
Variational Inference	7	Topological Data Analysis	1
Wigner-Ville Distribution	7	Word Embeddings	1

TABLE 17. Consolidated frequency of techniques related to fault prognosis and RUL after Step 4 of the review protocol (final corpus of 984 documents; Techniques Cited in More than 10 Documents are Highlighted in Bold).

Technique Name	Frequency	Technique Name	Frequency
Wavelet	**144**	Causal Inference	1
Long Short-Term Memory	**132**	Chained Sequential Memory	1
Convolutional Neural Network	**104**	Complete Ensemble Empirical Mode Decomposition with Adaptive Noise	1
Particle Filter	**91**	Conditional Generative Adversarial Networks	1
Support Vector Machine	**80**	Contextual Information Architecture	1
Principal Component Analysis	**78**	Continuous Deep Belief Network	1
Kalman Filter	**62**	CoxNet	1
Clustering	**53**	Cumulative Summation Features	1
Hidden Markov Models	**53**	Curve-Fitting Functions	1
Support Vector Regression	**50**	Deep Domain-Adversarial Regression Network	1
Anomaly Detection	**49**	Deep Gated Recurrent Neural Network	1
Recurrent Neural Network	**48**	Deep Graph Neural Networks	1
Transfer Learning	**44**	Deep Separable Convolutional Gated Recurrent Network	1
Autoencoder	**42**	Dimensionality Reduction Techniques	1
Memory Networks	**31**	Discrete Frechet Distance	1
Bayesian Network	**26**	Discrete Wavelet Decomposition	1
Brownian Motion	**26**	Double Exponential Smoothing	1
Knowledge-Based Expert System	**26**	Double-Layer Attention Mechanism	1
Self-Organizing Map	**26**	Dual Attention Mechanisms	1
Deep Neural Network	**25**	Dynamic Bayesian Network	1
Fuzzy Logic	**25**	Dynamic Time Warping	1
Gaussian Process Regression	**25**	Dynamic-inner Principal Component Analysis	1
Semi-Markov Models	**24**	Dynamic-Multilayer Graph Convolution Network	1
Deep Belief Network	**23**	Elastic Net	1
Gaussian Mixture Model	**21**	Ellipsoid-based Extended Kalman Filter	1
Generative Adversarial Networks	**21**	Elman Neural Network	1
Deep Learning	**20**	Encoder-Decoder Architectures	1
Extreme Learning Machines	**19**	Enhanced Principal Component Analysis	1
Relevance Vector Machine	**19**	Ensemble Learning	1
Deep Bidirectional Long Short-Term Memory Network	**17**	eXplainable Artificial Intelligence	1
Fourier Transform	**15**	Extreme Random Forest	1
Decision Tree	**14**	Feature Extraction	1
Logistic Regression	**14**	Feature-Transferred Prediction Network	1
Random Forest	**14**	Feedback Neural Network	1
Hidden Semi-Markov Models	**13**	Feedforward Neural Network	1
Auto Regressive Moving Average	**12**	Few-Shot Learning	1
Time Series Analysis	**11**	Fitting Interval Selection Method	1
Auto Regressive Integrated Moving Average	**10**	Frequency Energy Tracking	1
Linear Regression	**10**	Gaussian Process Classification	1
Stacking	**10**	Gradient Boosting Survival Analysis	1
Reinforcement Learning	9	Gradient-Boosted Trees	1
Short-Time Fourier Transform	9	Graph Based Models	1
Time Series Forecasting	9	Graph Embedding	1
Hilbert-Huang Transform	8	Graph-Based Models	1
Independent Component Analysis	8	Hilbert Marginal Spectrum	1
Kernel Density Estimation	8	Hilbert-Full-Vector Spectrum	1
K-Means	7	Hodrick Prescott Filter	1
Self-Attention Mechanisms	7	Hybrid Attention-based Convolutional Long Short-Term Memory	1
Attention Mechanisms	6	Inverse Gaussian Process	1
Bayesian techniques with Particle Filters	6	Isolation Forest	1
Stacked Autoencoder	6	Jordan Neural Network	1
Density-Based Spatial Clustering of Applications with Noise	5	Kernel Principal Component Analysis	1
Graph Convolutional Networks	5	Kernel Regression	1
Graph Neural Networks	5	Kernel Ridge Regression	1
K-Nearest Neighbors	5	Laguerre Filter	1
Static Bayesian Network	5	Least Squares Recurrent Auxiliary Classifier Generative Adversarial Network	1
Temporal Convolutional Networks	5	Lebesgue Sampling Long Short-Term Memory Network	1
Variational Inference	5	Light Convolutional Neural Network	1
Exponential Smoothing	4	Lightweight Bidirectional Long Short-Term Memory	1
Linear Discriminant Analysis	4	Lightweight Model Design	1
Ridge Regression	4	Linear Rectification Technology	1
State Space Models	4	Mahalanobis Distance	1
Transformer Models	4	Markov Random Field	1
Wigner-Ville Distribution	4	Maximum Mean Discrepancy	1
Bootstrap Aggregating	3	Metric Learning	1
Continuous Learning	3	Mixture Slow Feature Analysis	1

TABLE 17. *(Continued.)* **Consolidated frequency of techniques related to fault prognosis and RUL after Step 4 of the review protocol (final corpus of 984 documents; Techniques Cited in More than 10 Documents are Highlighted in Bold).**

Continuous Wavelet Transform	3	Monte Carlo Dropout	1
Deep Autoencoder	3	Monte Carlo Simulations	1
Deep Reinforcement Learning	3	Multi-Branch Improved Convolutional Network	1
Empirical Mode Decomposition	3	Multi-Feature Triggering	1
Extended Kalman Filter	3	Multilayer-Graph-Embedded Extreme Learning	1
Extreme Gradient Boosting	3	Multi-Model Data-Fusion-based Deep Transfer Learning	1
Gated Recurrent Unit	3	Multinomial Logistic Regression	1
Least Absolute Shrinkage and Selection Operator	3	Multi-Scale Convolutional Neural Network	1
Meta-Learning	3	Multi-Scale Deep Convolutional Neural Network	1
Multi-Layer Perceptron	3	Multi-scale Kernel Regression	1
Multiple Linear Regression	3	Multi-Source Adversarial Online Regression	1
Time Domain Analysis	3	Multi-Task Gaussian Process	1
Adaptive Resonance Theory	2	Multivariate Spatial-Temporal Graph Neural Network	1
Bayesian Inference	2	Neural Basis Expansion Analysis for Time Series	1
Broad Learning System	2	Neuro-Fuzzy	1
Convolutional Long Short-Term Memory	2	Nonlinear Autoregressive Neural Network	1
Deep Convolutional Neural Network	2	Novelty Detection	1
Deep Transfer Learning	2	Orthogonal Functions	1
Discrete Wavelet Transform	2	Outlier Detection	1
Domain Adaptation	2	Power Transform	1
Domain Generalization	2	Priori Knowledge-Enhanced Particle Filter	1
Exponentially Weighted Moving Average	2	Probabilistic Meta-Learning	1
Geometric Fractional Levy Stable Motion	2	Proportional Hazard Models	1
Kernel Methods	2	Quantum Machine Learning	1
Lebesgue Sampling	2	Random Walks	1
Local Outlier Factor	2	Read-First-Long Short-Term Memory	1
Markov Chain Monte Carlo	2	Recurrence Quantification Analysis	1
Matrix Factorization	2	Residual Hybrid Network	1
Model Pruning	2	Sample Entropy	1
Multi-Source Domain Adaptation	2	Savitzky-Golay Filter	1
Naive Bayes	2	Segmented Data Labels	1
Neural Architecture Search	2	Self-Adaptive Graph Convolutional Network	1
Non-Negative Matrix Factorization	2	Self-Adaptive Neural Networks	1
Particle Swarm Optimization	2	Self-Attention Monte Carlo Dropout	1
Simulated Annealing	2	Self-Supervised Learning	1
Statistical Pattern Recognition	2	Sequential Memory Networks	1
Survival Analysis	2	Shifting Kernel Particle Filter	1
T-Distributed Stochastic Neighbor Embedding	2	Siamese Long Short-Term Memory Network	1
Variational Autoencoders	2	Slow-Varying Dynamics-Assisted Temporal Capsule Network	1
Accumulative Self-Attention	1	Snapshot-Stacked Ensemble Deep Neural Network	1
Active Learning	1	Sparse Convolution	1
Adaptive Extended Kalman Filter	1	Spatial-Temporal Multi-Scale Graph Convolutional Neural Network	1
Adaptive Recognition Method	1	Spectral Clustering	1
Adversarial Machine Learning	1	Spiral Self-Attention Neural Network	1
Adversarial Out-domain Augmentation	1	Squeeze-and-Excitation Convolutional Neural Network	1
Adversarial Regression	1	Stacking Network Layers	1
Artificial Neural Networks	1	Stochastic Ordering	1
Bayesian Approximation	1	Support Vector Classification	1
Bayesian Deep Learning	1	Survival Random Forest	1
Bayesian Neural Networks	1	Temporal Attention Fusion Mechanism	1
Bayesian Optimization	1	Temporal Cascade Broad Learning System	1
Bayesian Transfer Learning	1	Tensor Representation	1
Bidirectional Gated Recurrent Unit	1	Time-Frequency Transformation Functions	1
Bidirectional Long Short-Term Memory Network	1	Time-Series Transformers	1
Bidirectional Recurrent Neural Network	1	Transductive Learning	1
Bootstrapping	1	Transfer Learning-based Bi-Directional Long Short-Term Memory	1
Box-Cox Transformation	1	Variational Mode Decomposition	1
Capsule Networks	1	Wasserstein Distance-based Multi-Scale Adversarial Domain Adaptation	1
Cascading Classifiers	1	Wavelet Neural Networks	1
Categorical Boosting	1	Word Embedding	1
Causal Augmented Convolution Network	1		

TABLE 18. Application context of the 40 most cited techniques in fault prognosis, including cluster assignment, prognosis stage coverage, and applications summaries.

Technique Name	Acronym	Cluster	Prognosis Stages	Application Summaries	References
Anomaly Detection	-	2	HS division	Anomaly detection is used in fault prognosis to identify structural changes in system behavior that signal transitions between HSs. Rather than relying on expert consensus than detecting isolated outliers, it captures temporal or contextual deviations in health indicators that mark the onset of degradation. These transitions can be used to trigger prognostic modeling and to segment the degradation process into distinct health stages. Techniques include statistical thresholding, dimensionality reduction with distance metrics, and machine learning methods such as CNNs, autoencoders, and ensemble classifiers – especially suited to scenarios with limited labels or low-frequency sensing.	[2], [36]
Auto Regressive Integrated Moving Average	ARIMA	2	DTP	ARIMA and its variants are used in fault prognosis to model and forecast degradation trends from time series data. These models capture autocorrelation and temporal dependencies to estimate the RUL, particularly under stationary or seasonally varying conditions. Extensions such as Seasonal ARIMA (SARIMA), Regression-augmented ARIMA (RegARIMA), and adaptive parameter tuning improve performance in complex environments, especially when degradation is influenced by cyclic loads, temperature variations, or other exogenous factors.	[37], [38], [39], [40]
Auto Regressive Moving Average	ARMA	2	HI construction	ARMA models are used in fault prognosis to model stationary time series that represent normal system behavior, enabling the extraction of residuals or transformed features for HI construction. They capture short-term temporal dependencies in degradation-related signals and are effective when the underlying dynamics follow linear autoregressive and moving average patterns. Commonly applied in vibration-based condition monitoring, ARMA serves as a baseline for identifying deviations from expected behavior and supporting early fault detection or HI derivation.	[25], [41]
Autoencoder	-	3	HI construction	Autoencoders are widely used in fault prognosis for unsupervised feature extraction and latent representation learning from sensory time-series data. They support HI construction by compressing high-dimensional signals into low-dimensional representations that capture degradation-related patterns. Variants such as Sparse Autoencoder, Variational Autoencoder (VAE), and Stacked Autoencoder enhance representation quality through sparsity, probabilistic modeling, or hierarchical encoding. These models are particularly effective in scenarios with limited labeled data or missing values and serve as a robust preprocessing stage for RUL estimation in mechanical systems.	[42], [43], [44], [45]
Bayesian Network	-	4	DTP	Bayesian Networks are used in fault prognosis to model probabilistic dependencies among system variables and degradation states. In this context, Dynamic Bayesian Networks (DBNs) propagate uncertainty across time by connecting state variables across multiple time steps, enabling the prediction of fault progression and RUL. Bayesian Belief Networks (BBNs), on the other hand, infer degradation trajectories by combining expert knowledge and observational data in a structured, probabilistic framework. These models are particularly effective when physical degradation distributions are unknown or labeled failure data is limited, as they allow for reasoning under uncertainty and integration of heterogeneous sources of information.	[46], [47]
Brownian Motion	-	2	DTP	Brownian motion models are used in fault prognosis to represent stochastic degradation paths under uncertainty. These models capture the random, continuous nature of wear processes and enable the prediction of degradation trajectories through probabilistic formulations. Fractional Brownian motion (fBm), an extension of classical Brownian motion, incorporates long-range temporal dependencies, making it suitable for systems with gradual or weak fault evolution. By modeling the degradation signal as a continuous-time stochastic process, these approaches support RUL estimation even in cases where the degradation is not easily segmented or linearly progressive.	[48]
Clustering	-	1	HS division; DTP	Clustering techniques are used in fault prognosis to segment health states, identify degradation patterns, and group assets or trajectories with similar failure behavior. These unsupervised methods support HS division by detecting regime shifts, latent degradation modes, or structural transitions in multivariate sensor or SCADA data. Clustering also enables DTP by organizing historical trajectories into reference groups, which are then used to infer the RUL of new units through similarity-based estimation. Methods such as k-means, DBSCAN, fuzzy c-means, and spectral clustering are commonly applied, often in combination with dimensionality reduction or distance metrics to capture meaningful degradation structure without requiring labeled failure data.	[5], [23], [3], [49], [50], [51], [52], [53]
Convolutional Neural Network	CNN	3	HI construction; DTP	CNNs are used in fault prognosis to extract spatial and temporal features from transformed sensor data, such as time-frequency maps, spectrograms, or wavelet encodings. These features support HI construction by capturing latent degradation patterns and serve as inputs for degradation modeling tasks, including RUL prediction. CNNs are commonly embedded within hybrid architectures, such as CNN-LSTM or CNN-BiLSTM, to leverage sequential	[7], [10], [23], [3], [42], [43], [44], [48], [54], [55],

TABLE 18. *(Continued.)* **Application context of the 40 most cited techniques in fault prognosis, including cluster assignment, prognosis stage coverage, and applications summaries.**

				modeling, and are extended to specialized forms like Graph Convolutional Networks (GCNs) for applications involving structured or relational data. In such hybrid models, CNNs typically perform spatial feature extraction from transformed data, while LSTM or related architectures model the sequential evolution of those features over time	[56], [57], [58], [59]
Decision Tree	-	1	DTP	Decision Trees are used in fault prognosis to classify health states, model degradation patterns, and support early fault prediction based on historical sensor data. These models are particularly effective in the HS division, where they segment operational data into discrete degradation categories without requiring extensive preprocessing. In some cases, Decision Trees are also employed to support HI construction by selecting relevant features or generating intermediate labels for weakly supervised learning. While less common for direct DTP, they are occasionally integrated into hybrid systems for RUL estimation or used to define interpretable rules for system behavior under evolving conditions. Their main advantage lies in transparency and ease of integration into industrial decision-making pipelines.	[8], [11], [27], [60], [61], [62], [63], [64]
Deep Belief Network	DBN	4	HI construction	DBNs are applied in fault prognosis for unsupervised feature extraction and latent health representation learning. Composed of stacked Restricted Boltzmann Machines (RBMs), DBNs learn hierarchical abstractions from sensor data and are commonly used for HI construction by compressing complex multivariate inputs into low-dimensional representations. In some cases, these features are further processed by regression models or deep hybrid architectures (e.g., DBN-LSTM) to predict RUL, enabling DTP. While less prevalent than CNNs or LSTMs in industrial applications, DBNs remain viable when prioritizes interpretability and unsupervised pretraining.	[21], [26], [54], [65]
Deep Bidirectional Long Short-Term Memory Network	BiLSTM	3	DTP	BiLSTM networks are widely used in fault prognosis to capture long-term temporal dependencies in both forward and backward directions, improving degradation pattern modeling. They are particularly effective in RUL prediction from sequential sensor data, especially in scenarios with multi-sensor inputs or non-stationary operational conditions. Recent studies frequently integrate BiLSTM with attention mechanisms to selectively emphasize relevant time steps or sensor channels, enhancing robustness and interpretability. Variants such as lightweight BiLSTM and transfer BiLSTM have also been proposed to address computational efficiency and domain adaptation.	[38], [55], [66], [67], [68], [69]
Deep Learning Architectures	-	3	HI construction; DTP	Deep Learning (DL) architectures are applied in fault prognosis to extract latent degradation patterns and predict future system behavior from complex sensor inputs. Unlike standard deep neural networks (DNNs), which operate on fixed-size structured data, DL architectures encompass specialized designs, such as convolutional neural networks (CNNs), recurrent networks (LSTMs, GRUs), temporal convolutional networks (TCNs), and hybrid models (e.g., CNN-LSTM, TCN-BiLSTM) that can model spatial and temporal dependencies directly. These models are used in both HI construction and DTP, supporting end-to-end learning from raw time series or multi-modal sensor streams. Advanced implementations may integrate attention mechanisms or transfer learning strategies to enhance performance under variable conditions. DL architectures are particularly well-suited for scenarios with high-dimensional, sequential, or heterogeneous data, where they often outperform conventional methods.	[14], [15], [26], [3], [42], [43], [67], [70], [71], [72], [73], [74]
Deep Neural Network	DNN	3	HI construction; DTP	DNNs, typically implemented as Multilayer Perceptrons (MLPs), are used in fault prognosis to construct HI and predict degradation trends by learning complex mappings from raw or preprocessed sensor signals. Their multi-layered structure enables the extraction of hierarchical features that correlate with system degradation. DNNs are especially useful when large labeled datasets are available or custom loss functions (e.g., divergence-based) guide the learning process toward monotonic or degradation-sensitive outputs.	[6], [23], [26], [42], [55], [63], [65], [71], [73]
Extreme Learning Machines	ELM	4	DTP	ELMs are used in fault prognosis for rapid regression modeling of RUL from extracted health indicators. Their single-layer feedforward architecture with randomly initialized hidden weights enables fast training and generalization in scenarios with limited computational resources. ELMs are often embedded in hybrid pipelines, such as graph-based or PCA-enhanced frameworks, to predict degradation trends from compressed feature representations.	[65], [75]
Fourier Transform	-	1	HI construction	Fourier Transform techniques, including Fast Fourier Transform (FFT), are used in fault prognosis to extract frequency-domain features from vibration or acoustic signals. These features reflect periodic behaviors and spectral shifts associated with system degradation. FT-based representations, such as spectrograms, are also commonly used as inputs for health indicator modeling or as preprocessing steps in data-driven prognostic pipelines.	[64], [76], [77], [78]
Fuzzy Logic	-	2	HI construction; DTP	Fuzzy logic systems are used in fault prognosis to model degradation under uncertainty, especially when crisp thresholds or deterministic models are not applicable. They support health indicator construction by mapping sensor inputs into linguistic states and enable DTP through rule-based reasoning or fuzzy	[7], [25], [60], [62], [79], [80], [81], [82]

TABLE 18. *(Continued.)* **Application context of the 40 most cited techniques in fault prognosis, including cluster assignment, prognosis stage coverage, and applications summaries.**

				transitions. Variants like Takagi–Sugeno models and neuro-fuzzy systems (e.g., ANFIS) allow for more accurate representation of complex system behaviors without requiring explicit time modeling. These systems are particularly suited for knowledge-rich environments or applications with scarce labeled data and ambiguous degradation patterns.	
Gaussian Mixture Model	GMM	1	HI construction	GMMs are used in fault prognosis to represent the probabilistic distribution of features extracted from sensor data. They support health indicator construction by modeling multivariate degradation patterns and estimating posterior probabilities over latent degradation states. GMMs are particularly useful in unsupervised settings and can be combined with divergence measures (e.g., Jensen-Rényi) to generate indicators that reflect system health evolution. These models are effective in scenarios with complex signal distributions and limited labeled data.	[23], [47], [83], [84], [85], [86], [87]
Gaussian Process Regression	GPR	6	DTP	GPR is used in fault prognosis to model degradation trajectories with quantified uncertainty. It enables RUL prediction by providing non-parametric, probabilistic estimates of system behavior over time. GPR is particularly suited for gradual degradation modeling and is often applied in scenarios where extrapolation and confidence bounds are essential for decision-making. Its ability to capture uncertainty makes it valuable in safety-critical or data-scarce prognostic settings.	[21], [22], [25], [65], [88], [89]
Generative Adversarial Networks	GAN	3	HI construction	GANs are used in fault prognosis to generate synthetic sensor data and augment underrepresented degradation patterns. This supports health indicator construction by addressing class imbalance and enhancing the diversity of training data in PHM tasks. Conditional and auxiliary GAN variants allow for the generation of class-specific samples, improving model robustness in scenarios with limited or imbalanced failure observations. These architectures are especially valuable in industrial contexts where labeled degradation data are scarce or costly to obtain.	[21], [26], [65], [90]
Hidden Markov Models	HMM	5	HS division; DTP	HMMs are used in fault prognosis to represent health states as latent degradation stages and to model transitions between them over time. HMMs support both health state division and DTP by capturing temporal dependencies in condition sequences. They are often combined with regression models or heuristic search to infer hidden states from feature trends and to estimate RUL probabilistically, even in scenarios with limited labeled data.	[23], [25], [83], [87]
Hidden Semi-Markov Models	HSMM	5	HS division; DTP	HSMMs are used in fault prognosis to model degradation processes where the duration of each latent health state follows an explicit, non-exponential distribution. HSMMs extend HMMs by allowing variable state dwell times, enabling more accurate modeling of real-world degradation behavior. They support health state division and probabilistic RUL prediction, and can be integrated into decision processes to optimize maintenance policies under uncertainty and cost constraints. These models are especially useful in systems with observable regime changes and where the timing of transitions carries diagnostic and prognostic significance.	[91], [92]
Kalman Filter	KF	2	HI construction; DTP	Kalman filters are used in fault prognosis to estimate latent degradation states and update model parameters in real-time based on sequential sensor observations. They support both HI construction and DTP by recursively correcting predictions with new measurements. Extended and nonlinear variants (e.g., EKF) enable RUL estimation in systems with non-Gaussian noise and nonlinear degradation dynamics. These filters are particularly useful in scenarios with limited data and provide uncertainty-aware tracking of the degradation process, making them well-suited for online prognosis in dynamic environments.	[25], [36], [80], [93]
Knowledge-Based Expert System	KBES	2	HS division; DTP	KBES are used in fault prognosis to simulate expert reasoning through rule-based inference, fuzzy logic, and component-specific knowledge bases. They support both HS division and DTP by encoding expert knowledge into interpretable decision structures. KBES are particularly effective in domains with limited data availability or where diagnostic expertise is well-established, and are often integrated with weak supervision techniques to expand labeling in semi-supervised settings. These systems offer a transparent alternative to black-box models and are well-suited for applications where interpretability and domain validation are critical.	[60], [62], [82]
Linear Regression	-	1	DTP	Linear regression models are used in fault prognosis to estimate RUL by mapping extracted health indicators to time-to-failure values. They are especially effective when degradation follows a linear trend and features have been dimensionally reduced. This entry includes variants such as Multiple Linear Regression (MLR), which allows the integration of multiple condition indicators into a single predictive model. Despite their simplicity, linear models remain valuable for interpretable and computationally efficient RUL estimation in systems where degradation evolves gradually and predictably.	[74], [94]

TABLE 18. *(Continued.)* **Application context of the 40 most cited techniques in fault prognosis, including cluster assignment, prognosis stage coverage, and applications summaries.**

Logistic Regression	-	2	DTP	Logistic regression is used in fault prognosis to estimate the probability of failure within a specified time horizon, based on features that capture system degradation. It is particularly effective for binary or multi-class classification tasks, such as predicting whether a component will fail within a given operational window. Logistic models are also integrated into probabilistic frameworks, such as Bayesian inference or particle filtering, for RUL estimation and risk-based maintenance decision-making. Due to its interpretability and low computational cost, logistic regression is suitable for real-time or embedded applications where explainability is a requirement.	[10], [11], [27], [63], [86]
Long Short-Term Memory	LSTM	3	DTP	LSTM networks are used in fault prognosis to model temporal dependencies in sensor data and estimate RUL. Their gated recurrent architecture enables the capture of long-range correlations in degradation sequences, supporting accurate trend prediction even under noisy or nonstationary conditions. LSTMs are applied in supervised frameworks, typically requiring fully labeled degradation trajectories or RUL annotations. Hybrid implementations may include attention mechanisms or physical regularization, but still rely on labeled data to guide training. Physics-guided variants incorporate domain knowledge as regularization terms, improving the physical consistency, monotonicity, and robustness of RUL predictions.	[5], [21], [23], [26], [39], [66], [67], [68], [71], [72], [95], [96], [97], [98], [99]
Memory Networks	-	3	HI construction; DTP	Although "Memory Networks" in the strict sense refers to architectures such as End-to-End Memory Networks [101] or Neural Turing Machines, in the fault prognosis literature the term is often used more broadly to encompass recurrent models with internal states that retain temporal information. In this broader sense, memory networks commonly refer to neural architectures with internal memory mechanisms, such as LSTM and RNN, which are capable of modeling temporal dependencies in degradation signals. These models support DTP by retaining and leveraging information over time, enabling more accurate RUL estimation and condition monitoring under sequential data streams. They are particularly effective when applied to time series from sensors and are often integrated with attention mechanisms, regularization strategies, or hybrid architectures to enhance performance and physical consistency. It is worth noting that LSTM, RNN, and CNN (also used within memory-enhanced hybrids) have dedicated entries due to their widespread adoption and more specifically defined roles in the reviewed literature.	[66], [67], [95], [96], [97]
Particle Filter	-	2	DTP	Particle filters are used in fault prognosis to estimate RUL by approximating the posterior distribution of degradation states in nonlinear, non-Gaussian systems. They operate by propagating a set of weighted samples (particles) through time, updating their likelihoods based on incoming observations. Particle filters are especially suited for uncertainty-aware prediction and real-time tracking of stochastic degradation trajectories. In prognostic applications, they are often combined with feature-driven or regression-based models to enhance robustness and compensate for model mismatch or noise in sensor data.	[7], [22], [86]
Principal Component Analysis	PCA	1	HI construction	PCA is used in fault prognosis to reduce the dimensionality of sensor or feature-extracted data, preserving the most informative components for HI construction. It enables degradation modeling by compressing statistical, spectral, or time-frequency features into a compact and orthogonal representation. PCA is often used to uncover latent degradation patterns, enhance signal-to-noise ratio, or support residual-based modeling within data-driven prognostic frameworks.	[23], [3], [45], [50], [58], [64], [75], [94], [101], [102]
Random Forest	-	1	HI construction; DTP	Random Forest is used in fault prognosis for both HI construction and DTP by aggregating decisions from multiple decision trees trained on bootstrapped feature subsets. In HI construction, RF enables mapping complex relationships between sensor-derived features and latent degradation states. For DTP, it serves as a robust regression model or as a base in survival analysis frameworks, such as Random Survival Forests, for estimating time-to-failure distributions. Random Forest handles high-dimensional, noisy, or partially censored data effectively, and supports uncertainty estimation through variance analysis across ensemble members. It is also integrated with feature extraction methods such as PCA, Tsfresh, or divergence-based metrics in hybrid prognostic pipelines.	[6], [12], [23], [79], [85], [102], [103]
Recurrent Neural Network	RNN	3	HI construction; DTP	RNNs are used in fault prognosis to model temporal dependencies in sensor data and predict future degradation behavior. In HI construction, RNNs learn compressed representations of system health directly from raw or preprocessed time series. For DTP, they are applied to forecast RUL based on sequential sensor observations. Variants such as Gated Recurrent Units (GRU), Bidirectional GRU (BiGRU), and NARX networks improve modeling robustness, enabling both supervised and hybrid implementations. RNNs are also integrated into multitask learning setups and augmented with attention mechanisms to capture long-range temporal patterns, enhancing their effectiveness in complex prognostic pipelines.	[23], [44], [72], [95], [104]
Relevance Vector Machine	RVM	2	DTP	RVMs are used in fault prognosis for probabilistic regression of degradation trends and RUL. As a sparse Bayesian alternative to Support Vector Machines,	[48], [86]

TABLE 18. *(Continued.)* **Application context of the 40 most cited techniques in fault prognosis, including cluster assignment, prognosis stage coverage, and applications summaries.**

				RVMs provide probabilistic outputs with automatic relevance determination, often yielding more compact models with fewer relevance vectors. Their ability to quantify uncertainty in predictions makes them suitable for applications requiring confidence-aware RUL estimates. RVMs are commonly applied in conjunction with entropy-based indicators or dimensionality reduction techniques to improve robustness and computational efficiency.	
Self-Organizing Map	SOM	1	HI construction	SOMs are used in fault prognosis to perform unsupervised clustering and visualization of health states based on sensor data. They support HI construction by projecting high-dimensional features into a low-dimensional topological map, enabling the detection and tracking of degradation patterns. SOMs have been applied to derive confidence values (CV) or Bayesian inference-based probabilities (BIP), which quantify machine health in a bounded scale (e.g., [0,1]). These probabilistic indicators are especially useful when labeled failure data are scarce and can serve as robust health indicators for subsequent DTP models.	[86], [88]
Semi-Markov Models	SMM	5	HS division; DTP	SMM are used in fault prognosis to represent system degradation through transitions among discrete health states, with explicitly modeled sojourn times governed by non-exponential distributions. By relaxing the memoryless property of Markov models, SMM enables more accurate modeling of failure dynamics under variable operating conditions. They support both HS division, by structuring multi-state degradation processes, and DTP, through analytical or numerical estimation of reliability functions, first-passage times, and expected RUL. SMM are particularly suitable when external factors influence the timing and likelihood of transitions between degradation stages	[105], [106]
Stacking	-	4	HS division; DTP	Stacking is applied in fault prognosis by combining feature extractors (e.g., CNNs, autoencoders) with temporal models (e.g., LSTM) in a fixed, end-to-end supervised architecture. Trained on labeled HI or RUL data, it captures both spatial and temporal degradation patterns. When configured for discrete health state classification, stacking supports HS division; when configured for continuous degradation modeling or RUL estimation, it supports degradation trend prediction (DTP). This approach improves accuracy when no single model suffices and is suited for fully labeled, data-rich scenarios.	[59]
Support Vector Machine	SVM	1	HS division; DTP	SVMs are applied in fault prognosis to classify discrete health states (HS division) and, in some cases, to model degradation trends via Support Vector Regression (SVR). They are especially effective when condition features are extracted from time-, frequency-, or order-domain vibration signals. Their margin-based decision function enables robust classification in high-dimensional spaces, making them suitable for segmenting health conditions prior to RUL estimation. In trend prediction tasks, SVR variants are often used with handcrafted features or fused indicators. SVMs are also commonly used as benchmark models in condition monitoring frameworks.	[7], [63], [65], [76], [78], [107], [108]
Support Vector Regression	SVR	1	DTP	SVR is applied in fault prognosis to model degradation trends and estimate RUL from features extracted in the time, frequency, or statistical domains. As a margin-based regression technique, it enables robust function approximation even with limited or noisy data. Its ability to generalize from sparse observations makes it especially effective in condition monitoring scenarios with scarce failure records. Variants such as Least Squares SVR (LSSVR) and hybrid approaches combine SVR with feature selection or optimization algorithms (e.g., Particle Filter, Evolutionary Computation) to improve prediction accuracy and model stability.	[7], [21], [63], [64], [65], [85], [87], [108]
Time Series Analysis	TSA	2	HI construction; DTP	Time series analysis is applied in fault prognosis to extract health indicators and model degradation trends from temporally ordered sensor data. In the HI construction stage, filtering techniques (e.g., autoregressive modeling, wavelet decomposition, moving averages) are used to isolate degradation-relevant features from noise and transient effects. For degradation trend prediction, statistical models such as AR, ARMA, ARIMA, and nonlinear extensions like NARX are employed to forecast future system behavior or Remaining Useful Life (RUL). These techniques allow for the detection of trends, seasonality, and anomalies, and are widely used in predictive maintenance frameworks across diverse industrial applications.	[22], [28], [97], [103], [104], [107], [109], [110]
Transfer Learning	-	3	DTP	Transfer learning is used in fault prognosis to enable DTP under domain shift, adapting models trained on source domains to new target domains with limited or unlabeled data. It transfers feature representations or model parameters across operating conditions, system configurations, or datasets. Common approaches include parameter-based fine-tuning, domain adaptation techniques such as MK-MMD, co-training with unlabeled data, and simulation-to-reality transfer using synthetic data. These methods are typically implemented within deep learning architectures (e.g., CNN, LSTM, BiGRU) to enhance model generalization and reduce labeling effort in industrial prognostic systems.	[21], [45], [59], [69], [71], [74]
Wavelet	-	1	HI construction; DTP	Wavelet transforms are used in fault prognosis to decompose sensor signals into time-frequency components, enabling the extraction of degradation-sensitive	[41], [52], [56], [88],

TABLE 18. *(Continued.)* **Application context of the 40 most cited techniques in fault prognosis, including cluster assignment, prognosis stage coverage, and applications summaries.**

	features. In the HI construction stage, techniques such as Discrete Wavelet Transform (DWT), Wavelet Packet Transform (WPT), and Continuous Wavelet Transform (CWT) are employed to isolate informative frequency bands from non-stationary vibration or current signals. For DTP, Wavelet-based Neural Networks (WNNs) and multi-wavelet coefficient fusion methods integrate spectral decomposition with adaptive modeling, supporting accurate estimation of RUL. These approaches enhance robustness under variable operational conditions and are frequently applied in rotating machinery, electric motors, and electronic systems.	[104], [108], [109], [110], [112], [113]

Second, by adopting the TOPSIS method, the framework ensures a transparent and replicable ranking mechanism. The use of tunable weights and discrete utility functions allows practitioners to model industrial preferences, such as simplicity, transparency, or scalability, without relying on opaque optimization techniques or requiring advanced MCDM expertise.

Sensitivity tests confirmed the framework's robustness: across variations in data conditions, evaluation preferences, and ranking consistency, the outcomes remained stable, interpretable, and coherent with the problem constraints.

However, although robustness was evaluated through multiple sensitivity and consistency analyses, further validation through comparative experiments in different industrial contexts and applications is left as a direction for future research. Such extensions would complement the present work by reinforcing its cross-domain applicability and strengthening confidence in practical deployment.

Crucially, the framework does not replace practitioner judgment, but it reinforces it. The inclusion of a post-selection feasibility assessment ensures that final decisions account for real-world constraints such as infrastructure, organizational readiness, and regulatory requirements. This safeguard proved decisive in the case study, where Transfer Learning, despite its top score, was replaced by ARIMA due to deployment limitations. Rather than prescribing a single solution, the framework frequently highlights families of suitable techniques, as seen in the close ranking of Autoencoder and PCA for HI construction. This supports more nuanced decision-making, where final choices may reflect algorithmic complementarity, domain expertise, or empirical testing.

Although the case study focused on a power substation, the challenges it exemplifies, such as the absence of labeled failure events, the need for interpretable models to support maintenance planning, and operational constraints regarding automation and model complexity, are widely observed in other industrial domains, including manufacturing, transportation, and process industries. Since the framework is grounded in abstract, domain-independent attributes and supports customizable evaluation criteria, it remains adaptable to a broad range of operational contexts, reinforcing its domain-agnostic nature and positioning it as a replicable decision-support tool across sectors.

This adaptability extends beyond application domains. While this study focuses on fault prognosis, the same two-phase decision logic could be adapted to fault detection and diagnosis by adjusting the selection parameters and incorporating task-specific foundational studies. For detection, emphasis would shift toward unsupervised anomaly recognition techniques, whereas diagnosis would benefit from interpretable classification models capable of identifying failure causes. Such extensions would enable unified model selection across all stages of condition monitoring, i.e., detection, diagnosis, and prognosis, under a consistent methodological approach.

The framework's flexibility and extensibility also apply to the ranking method itself. Although the current implementation employs TOPSIS for its transparency and simplicity, other multi-criteria techniques, such as PROMETHEE, could be integrated to enable more granular preference modeling, particularly in collaborative or multi-stakeholder contexts.

In summary, the proposed framework provides a transparent, adaptable, and scalable tool for guiding the selection of ML techniques in fault prognosis. It moves beyond benchmark- and accuracy-driven comparisons, enabling a reasoned synthesis of data characteristics, modeling constraints, and deployment requirements. While developed in the context of electrical power infrastructure, its principles generalize to a wide range of industrial systems where reliability, interpretability, and feasibility are critical for operational success.

APPENDIX

See Tables 15–19.

TABLE 19. Taxonomy of machine learning techniques according to the 12 key selection attributes defined in the present study.

Technique Name	LR	DT*	DA	AL	TR	GS	IN	TE	MC	AGC	AA	AD	References
Anomaly Detection	Pr	St	Mr	Sa	Lm	Sp	2	2	2	2	Anomaly Detection	Manufacturing; Electronics	[2], [36], [41], [114], [115]
Auto Regressive Integrated Moving Average	Pr	Sq	Mr	Sa	Lm	Md	3	3	1	1	Statistical Time-Series	Manufacturing; Transportation	[2], [25], [37], [39], [40], [83], [107], [110]
Auto Regressive Moving Average	Pr	Sq	Mr	Sa	Lm	Md	3	3	1	1	Statistical Time-Series	Energy; Manufacturing	[25], [40], [41], [77], [112]
Autoencoder	Nr	St; Sq	Mr	Sa	Ad	Md	2	1	2	3	Neural Networks	Manufacturing; Transportation; Energy	[10], [26], [42], [43], [44], [45], [54], [65], [83], [113], [116]
Bayesian Network	Fr	St	Sc	Mn	Ad	Md	3	3	2	2	Bayesian	Transportation	[46], [47]
Brownian Motion	Nr	Sq	Mr	Mn	Lm	Sp	2	2	2	1	Stochastic Processes	Energy	[48]
Clustering	Nr	St	Sc	Fa	Ad	Md	2	2	2	2	Clustering	Manufacturing; Energy; Electronics	[5], [23], [3], [41], [49], [50], [51], [52], [53], [117]
Convolutional Neural Network	Fr	St; Sq	Rr	Sa	Ad	Md	1	1	3	3	Neural Networks	Manufacturing; Energy; Transportation	[5], [7], [8], [10], [21], [23], [26], [3], [42], [43], [44], [48], [54], [55], [56], [57], [58], [59], [65], [73], [74], [78], [83], [113], [116]
Decision Tree	Fr	St	Mr	Sa	Ad	Md	3	3	1	2	Decision Trees	Energy; Manufacturing	[5], [7], [8], [10], [11], [24], [27], [3], [60], [61], [62], [63], [64], [65], [79]
Deep Belief Network	Fr	St	Rr	Sa	Ad	Sp	1	1	3	3	Neural Networks	Manufacturing; Energy	[21], [26], [54], [65], [116], [118]
Deep Bidirectional Long Short-Term Memory Network	Fr	Sq	Rr	Sa	Ad	Md	1	1	3	3	Neural Networks	Manufacturing	[38], [55], [66], [67], [68], [69], [70], [111]
Deep Learning Architectures	Fr	Sq	Rr	Sa	Ad	Md	1	1	3	3	Neural Networks	Manufacturing; Energy	[14], [15], [26], [3], [42], [43], [54], [67], [70], [71], [72], [73], [74], [83], [116], [118], [119]
Deep Neural Network	Fr	St	Rr	Sa	Ad	Md	1	1	3	3	Neural Networks	Manufacturing; Transportation; Energy	[6], [23], [26], [42], [55], [63], [65], [71], [73]
Extreme Learning Machines	Fr	St	Mr	Sa	Lm	Sp	2	2	2	2	Neural Networks	Manufacturing	[65], [75]
Fourier Transform	Nr	St	Mr	Mn	Ad	Md	3	2	1	1	Signal Processing	Manufacturing; Energy	[64], [76], [77], [78]
Fuzzy Logic	Nr	St	Mr	Mn	Ad	Md	3	3	2	2	Fuzzy Logic	Manufacturing	[7], [23], [25], [60], [62], [79], [80], [81], [82], [120]
Gaussian Mixture Model	Nr	St	Mr	Sa	Ad	Md	2	2	2	2	Bayesian	Manufacturing	[23], [47], [83], [84], [85], [86], [87]
Gaussian Process Regression	Fr	St	Mr	Mn	Lm	Sp	2	2	2	2	Bayesian Methods	Manufacturing; Energy	[21], [22], [25], [65], [88], [89]
Generative Adversarial Networks	Fr	St	Rr	Sa	Lm	Sp	1	1	3	3	Neural Networks	Manufacturing	[21], [26], [65], [90]
Hidden Markov Models	Fr	Sq	Mr	Sa	Ad	Md	2	1	2	2	Bayesian	Manufacturing	[23], [25], [83], [87]
Hidden Semi-Markov Models	Fr	Sq	Mr	Mn	Lm	Sp	2	2	2	2	Probabilistic Modeling	Transportation; Energy	[91], [92]
Kalman Filter	Pr	Sq	Mr	Sa	Ad	Md	3	3	2	2	Bayesian	Transportation	[25], [36], [80], [93], [121]

TABLE 19. *(Continued.)* **Taxonomy of machine learning techniques according to the 12 key selection attributes defined in the present study.**

Knowledge-Based Expert System	Nr	St	Mr	Mn	Lm	Sp	3	3	1	1	Knowledge-Based Systems	Energy	[60], [62], [82]
Linear Regression	Fr	St	Sc	Sa	Ad	Gp	3	3	1	1	Regression	Energy	[22], [74], [94]
Logistic Regression	Fr	St	Mr	Sa	Ad	Gp	3	3	1	1	Statistical Models	Energy; Manufacturing	[10], [11], [27], [63], [83], [86]
Long Short-Term Memory	Fr	Sq	Rr	Sa	Ad	Md	1	1	3	3	Neural Networks	Manufacturing; Transportation; Energy	[5], [7], [10], [21], [23], [26], [39], [43], [44], [56], [58], [59], [63], [66], [67], [68], [71], [72], [83], [95], [96], [97], [98], [99], [111], [116], [122]
Memory Networks	Fr	Sq	Rr	Sa	Lm	Sp	1	1	3	3	Neural Networks	Manufacturing; Transportation	[66], [68], [95], [96], [97]
Particle Filter	Fr	Sq	Mr	Sa	Ad	Md	2	2	2	2	Bayesian	Energy	[7], [22], [25], [86]
Principal Component Analysis	Nr	St	Sc	Sa	Ht	Gp	3	3	1	2	Dimensionality Reduction	Manufacturing, Industrial, Energy; Transportation	[5], [10], [23], [3], [45], [50], [58], [65], [75], [77], [83], [94], [101], [102]
Random Forest	Fr	St	Mr	Sa	Ad	Md	2	2	2	3	Decision Trees	Energy, Manufacturing	[5], [6], [8], [10], [12], [19], [23], [24], [27], [65], [79], [83], [85], [102], [103]
Recurrent Neural Network	Fr	Sq	Rr	Sa	Ad	Md	1	1	3	3	Neural Networks	Manufacturing; Transportation	[5], [10], [23], [26], [44], [65], [72], [73], [83], [90], [95], [104]
Relevance Vector Machine	Fr	St	Mr	Sa	Ad	Md	2	2	2	2	Bayesian	Manufacturing	[79], [86], [103]
Self-Organizing Map	Nr	St	Sc	Sa	Ad	Md	2	2	2	2	Clustering	Manufacturing	[79], [86], [88]
Semi-Markov Models	Fr	Sq	Mr	Mn	Lm	Sp	2	2	2	2	Probabilistic Modeling	Transportation; Energy	[105], [106]
Stacking	Fr	St	Mr	Sa	Ad	Md	2	2	3	3	Ensemble Learning	Manufacturing; Transportation	[59]
Support Vector Machine	Fr	St; Sq	Mr	Sa	Ad	Md	2	2	2	2	Support Vector Machines	Energy, Manufacturing	[5], [7], [8], [10], [11], [19], [21], [24], [26], [27], [51], [63], [65], [76], [78], [79], [83], [95], [107], [108], [113], [123]
Support Vector Regression	Fr	St	Mr	Sa	Ad	Md	2	2	2	2	Support Vector Machines	Energy	[7], [10], [21], [23], [25], [51], [63], [64], [65], [74], [83], [85], [87], [97], [108]
Time Series Analysis	Fr	Sq	Mr	Sa	Ad	Md	2	2	2	2	Time Series Analysis	Energy; Manufacturing	[10], [19], [22], [25], [3], [83], [97], [103], [104], [107], [109], [110]
Transfer Learning	Pr	St	Mr	Sa	Ad	Gp	2	2	2	3	Transfer Methods	Manufacturing; Transportation	[21], [45], [59], [69], [71], [74]
Wavelet	Nr	Sq	Mr	Sa	Ad	Md	2	2	1	2	Signal Processing Techniques	Manufacturing; Energy; Electronics	[5], [42], [52], [56], [65], [77], [88], [104], [108], [109], [111], [112], [113], [123], [124], [125], [126], [127], [128], [129]

*While all value categories were defined to support broader generalization of the framework, certain values, such as "Un" (Unstructured) and "Hy" (Hybrid) under the Data Type (DT) attribute, do not appear in the final mapping of the 40 most frequent techniques due to lack of representation in the reviewed corpus.

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